Comparing Mutual Information in Finite Samples: Wald and Expanded Welch–Satterthwaite Tests

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A thesis submitted in partial fulfilment of the requirements for the degree of

Master of Computer Science

School of Computer Science
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Australia

2026

Abstract

Mutual information (MI) measures the average reduction in uncertainty about one variable obtained by observing another. This thesis compares MI between the same two categorical variables in two independently sampled populations whose joint probability tables can differ even when their MI values are equal. The Normal Wald test uses a normal approximation that can be inaccurate in small or sparse samples.

The thesis develops an Expanded Welch–Satterthwaite test by deriving how changing one observation affects the estimated MI variance. By accounting for changes to cell probabilities and their row and column totals, the calculation supplies the degrees of freedom for a Student t distribution while retaining Wald's estimated difference and standard error.

Three controlled studies examine false positives, degrees of freedom and statistical power to detect MI differences across $2 \times 2, 4 \times 4, 8 \times 8$ and 16×16 tables and three margin regimes. Supporting checks vary construction and sample allocation. Each of the 240 settings uses 20,000 sampled table pairs. Diagnostics assess the shared MI estimates and the predicted sampling variability of the estimated MI variance.

Expanded Welch brings false-positive rates closer to 5% in some small-sample settings, but makes already-conservative rates lower still. Near independence, small estimated degrees of freedom reinforce this conservatism. In 16×16 tables with different marginal probabilities at $n = 1{,}000$, unequal MI bias shifts the estimated difference away from zero. False-positive rates are 0.2522 for Wald and 0.2402 for Expanded Welch, with every calculation valid. At small nonzero differences, this bias can also reverse the estimated ordering among rejections. The supporting construction check reproduces both failures. Larger samples substantially reduce bias, improve power and bring the two methods closer together.

The evaluation does not support Expanded Welch as a general replacement for Normal Wald, because its larger rejection cutoff cannot correct errors in the shared MI difference or standard error.

Acknowledgements

I am grateful to my supervisor, Associate Professor Joseph T. Lizier, for guidance throughout this research. I also thank the colleagues, friends, and family members whose support made this work possible.

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Chapter 1

Introduction

Mutual information (MI) measures the average reduction in uncertainty about one variable obtained by observing another (Bossomaier et al., 2016). For example, a categorical input feature X may carry different amounts of information about a class label Y in two populations. This thesis develops and evaluates a statistical test for that comparison using two independently collected samples.

Let P and Q be the joint probability distributions of X and Y in the two populations. The null and alternative hypotheses are

$$H_0 : I(P) = I(Q), \qquad H_1 : I(P) \neq I(Q). \tag{1.1}$$

Under H_0, the populations have equal MI. However, this does not mean that their joint probability tables are the same, as two populations can have equal MI while having different probability tables.

In practice, MI is estimated from observed data. Two samples can produce different MI estimates even when the population MI values are equal, because their observed counts vary from sample to sample. A statistical test must account for this sampling variation before an estimated difference can be interpreted as evidence of a population difference.

1.1 Why the comparison is difficult

The Normal Wald test (Wald, 1943; van der Vaart, 1998) divides the estimated MI difference by its standard error and compares the resulting statistic with a standard normal distribution. At a 5% significance level, the intended false-positive rate is 5% when the population MI values are equal. The normal approximation can be

inaccurate in small or sparse samples, causing the test to reject equal MI too often or too rarely.

Both the MI difference and its standard error are estimated from count tables, whose cells record how often each pair of categories occurs. MI depends on the cell probabilities and their row and column totals. When samples are small and rare combinations of categories occur only a few times or not at all, the resulting probability estimates can distort both the MI difference and its estimated uncertainty.

The proposed Expanded Welch–Satterthwaite test (Expanded Welch) retains Wald's estimated MI difference and standard error but uses a Student t distribution instead of the standard normal distribution. Choosing its degrees of freedom requires estimating how much the MI variance estimates change from sample to sample. For MI, this calculation must account for the fact that changing a cell probability also changes its row and column totals and the MI values calculated from them. This thesis derives how these probability changes affect the estimated MI variance and uses that calculation to obtain degrees of freedom for Expanded Welch.

The Student t distribution gives Expanded Welch a larger rejection cutoff than Normal Wald, so it rejects less often. This can reduce false positives, but it can also reduce statistical power: the probability of rejecting equal MI when the population MI values genuinely differ. The evaluation therefore examines both outcomes, together with how often each method returns a usable p-value, to determine whether the larger cutoff offers a useful improvement.

1.2 Research questions

The thesis addresses two research questions:

1. How can the sampling variability of the estimated MI variance determine MI-specific degrees of freedom for Expanded Welch?

2. Under what conditions does Expanded Welch improve false-positive rates relative to Normal Wald, and what are the costs in power, usable results and computation?

An analytical derivation addresses the first question, while controlled simulations with known population MI address the second by varying sample size, table size, category frequencies and population MI. Estimation diagnostics explain how bias and uncertainty affect the test results.

1.3 Contributions to the field

1. **Mathematical contribution.** The thesis derives how changing one observation affects the estimated MI variance, including the resulting changes in the row and column totals and in pointwise MI.

2. **Methodological contribution.** Expanded Welch uses this variance sensitivity to choose the degrees of freedom for a two-sample test using a Student t distribution.

3. **Empirical contribution.** Controlled simulations examine how sample size, table size, category frequencies and population MI affect false-positive rates, power and the frequency of usable results. Supporting diagnostics assess errors in the MI difference and its standard error, and directly check the predicted variability of the MI variance estimate. Additional supplementary checks look at further population constructions and sample allocation, while a timing comparison measures computational cost.

4. **Practical contribution.** The evaluation identifies when Expanded Welch brings false-positive rates closer to the intended 5% level than Normal Wald, and when estimation errors make both tests unreliable.

1.4 Scope and organisation

The scope of this thesis is the comparison of MI between two populations using categorical data. Each sample is summarised in a count table, and both tables use the same fixed row and column categories. The two samples are independent, and observations within each sample are also independent. For the main derivation, both populations are assumed to have positive MI and a positive probability in every cell. The case of zero MI, where X and Y are independent, is considered separately. Normal Wald and Expanded Welch are compared using two-sided tests at the 5% significance level.

The organisation of the thesis follows the development, evaluation and interpretation of the two-sample MI test. Chapter 2 reviews the background and literature. Chapters 3 and 4 define the shared statistic and derive Expanded Welch. Chapter 5 describes the experiment, and Chapter 6 discusses the results. Chapter 6 also illustrates both tests through a real-world example using an online-shopping dataset. Chapters 7 and 8 discuss the implications, limitations and conclusions. The appendices provide supporting calculations, experimental settings and additional results.

Chapter 2

Background and Literature Review

Testing for equal MI in two populations requires an estimate from each sample and its sampling variance. This chapter develops the MI estimator, derives its variance and reviews tests for equal population MI.

2.1 Information-theoretic foundations

2.1.1 Entropy and information content

For a discrete random variable X with probability mass function $p_X(x)$, the information content of outcome x is $-\log p_X(x)$. Rare outcomes have larger information content because they are less expected. Using natural logarithms gives information in nats. The Shannon entropy is the expected information content,

$$H(X) = -\sum_x p_X(x) \log p_X(x). \tag{2.1}$$

Entropy summarises the uncertainty of one random variable before its value is observed (Shannon, 1948).

For a pair (X, Y), the joint entropy $H(X, Y)$ is defined from the joint probabilities in the same way. The conditional entropy $H(X \mid Y) = H(X, Y) - H(Y)$ measures the uncertainty remaining in X after observing Y.

2.1.2 Pointwise and average mutual information

Let $P = (p_{ij})$ be the joint distribution of categorical variables $X \in \{1, \ldots, r\}$ and $Y \in \{1, \ldots, c\}$. The numbers of categories, r and c, are their alphabet sizes. Write

$$p_{i+} = \sum_{j=1}^{c} p_{ij}, \qquad p_{+j} = \sum_{i=1}^{r} p_{ij} \tag{2.2}$$

for the marginal probabilities, or row and column margins. The pointwise mutual information (also called local MI) associated with cell (i, j) is

$$\ell_P(i, j) = \log\left(\frac{p_{ij}}{p_{i+}p_{+j}}\right). \tag{2.3}$$

The denominator is the cell probability under independence with the same margins. A positive value means that the cell occurs more often than independence predicts, whereas a negative value means that it occurs less often.

Mutual information is the probability-weighted average of these pointwise values (Shannon, 1948; Bossomaier et al., 2016):

$$I(P) = \sum_{i=1}^{r}\sum_{j=1}^{c} p_{ij}\ell_P(i, j). \tag{2.4}$$

Here the summations are nested: for each row i, the inner sum runs over all columns j. Equivalently, the expression sums once over every cell in the table and may be written with the shorthand $\sum_{i,j}$.

Because P is the joint distribution of (X, Y), we also write $I(P)$ as $I(X; Y)$. Its entropy form is

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \tag{2.5}$$

MI is therefore the average reduction in uncertainty about X obtained by observing Y, or equivalently about Y obtained by observing X. It is nonnegative and equals zero exactly when $p_{ij} = p_{i+}p_{+j}$ for every cell, including zero-probability cells (Bossomaier et al., 2016).

Although MI measures the overall departure from independence, it does not specify the shape of association. Two populations can have the same MI while assigning probability to different cells, having different margins, or exhibiting different association patterns. Equality of MI is therefore weaker than equality of distributions. The null hypothesis requires equal MI, not identical cell probabilities in the two population tables.

The logarithm base determines the reporting unit. This thesis uses natural logarithms, so MI is measured in nats; dividing by $\log 2$ converts to bits. This conversion rescales MI and its standard error together but leaves the degrees of freedom unchanged, so the test result is identical in either unit.

2.2 Estimating discrete mutual information

Suppose a sample of size n produces counts N_{ij}. The empirical cell probabilities are

$$\widehat{p}_{ij} = \frac{N_{ij}}{n}. \tag{2.6}$$

The plug-in estimator inserts these observed relative frequencies into Equation (2.4):

$$\widehat{I}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij} \log\left(\frac{\widehat{p}_{ij}}{\widehat{p}_{i+}\widehat{p}_{+j}}\right). \tag{2.7}$$

The zero-cell convention follows from $\lim_{x\downarrow 0} x \log x = 0$, so empty cells contribute no term to the sum. When a positive-probability cell is unobserved, the empirical table assigns it zero probability, contributing to sparse-sample estimation error.

Under multinomial sampling, the cell counts vary between samples but must sum to n. An extra observation in one cell therefore leaves fewer observations for the others. For a fixed set of categories, the plug-in estimate approaches population MI as the sample grows, but its finite-sample bias and sampling variance depend on the complete probability table. In particular, rare cells can change both even when the categories and sample size are unchanged (Paninski, 2003).

For a fixed positive $r \times c$ support, the large-sample expansion of the plug-in estimator's bias has a positive leading term:

$$\text{Bias}\{\widehat{I}(P)\} = \frac{(r-1)(c-1)}{2n} + O(n^{-2}). \tag{2.8}$$

The leading term grows with table size and shrinks as the sample grows, although it does not describe every small or sparse sample. The result is closely related to Miller's entropy correction (Miller, 1955), and the same histogram-MI term appears in Moddemeijer (1989). Sparse sampling can produce further error beyond this leading approximation (Paninski, 2003).

Bias matters directly when comparing MI estimates from two populations. If $I(P) = I(Q)$ but $n_P \neq n_Q$, the uncorrected difference has leading bias

$$\frac{(r-1)(c-1)}{2}\left(\frac{1}{n_P} - \frac{1}{n_Q}\right). \tag{2.9}$$

For a 2×2 table with $n_P = 50$ and $n_Q = 100$, Equation (2.9) gives 0.005 nats despite equal population MI values. Chapter 3 corrects this sample-size effect by subtracting $(r-1)(c-1)/(2n)$ from each estimate.

Estimation beyond the leading bias correction.

Data are undersampled when too few observations are available to estimate the joint probabilities reliably. Rare category combinations can then be poorly represented or absent from the sample. Panzeri and Treves (1996) derive analytical bias corrections that account for the categories with nonzero population probability, but these remain large-sample approximations. Paninski (2003) shows that substantial error can remain after standard corrections, including the jackknife, when many categories are poorly sampled. Chao and Shen (2003) instead adjust entropy for categories missing from the sample. Their method uses sample coverage, the total population probability of the categories that have been observed, and accounts for the different probabilities of discovering common and rare categories.

Bayesian estimators model the unknown probabilities rather than treating observed proportions as the complete population. Nemenman et al. (2002) combine priors, or probability weights on possible populations, to avoid favouring a narrow range of entropy values before observing the data. This is the Nemenman–Shafee–Bialek (NSB) estimator. More recently, Piga et al. (2024) estimate the strength of a Dirichlet prior from the observed counts. This strength controls how much the probability estimates are drawn towards a uniform distribution, including for unobserved categories.

Entropy estimators can also be combined to estimate MI through $I(X;Y) = H(X) + H(Y) - H(X,Y)$, as in Equation (2.5). Archer et al. (2013) study this route alongside estimators that model the joint probabilities directly. Combining three separately constructed Bayesian entropy estimates need not correspond to one joint probability model and can produce negative MI estimates. Nevertheless, these combined estimators generally outperform the authors' fully Bayesian MI estimator in their tested examples.

The sampling pattern can also suggest a more targeted estimator. Hernández and Samengo (2019) consider a well-sampled, low-entropy variable paired with a variable whose many categories are poorly sampled. They use the well-estimated marginal

and model the conditional probabilities to estimate MI, rather than estimating three entropies separately. Their results also show that combining Bayesian entropy estimates can retain bias under severe undersampling. Estimation errors need not cancel when MI estimates from two distinct populations are subtracted. The two-sample problem therefore requires both a small bias in the estimated difference and a standard error that describes its sampling variation.

2.3 Sampling variance of plug-in MI

Comparing two MI estimates requires knowing how much each varies across repeated samples. We derive the large-sample approximation

$$\mathrm{Var}\{\widehat{I}(P)\} \approx \frac{V(P)}{n}. \tag{2.10}$$

Here $V(P)$, defined below, measures how much the contribution of one observation varies across the population. The derivation assumes that the population cell probabilities are positive, so the logarithms and their derivatives are well defined. This first-order variance is established for histogram and discrete MI (Moddemeijer, 1989; Brillinger, 2004).

The argument proceeds in three steps. First, sampling changes the empirical cell probabilities. Second, a first-order Taylor expansion translates those probability errors into an MI error. Third, that error can be written as a sample average, whose variance is straightforward to calculate.

Step 1: From probability errors to MI error.

Using the entropy identity in Equation (2.5), MI can be written directly in terms of the joint and marginal probabilities:

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_{i} p_{i+} \log p_{i+} - \sum_{j} p_{+j} \log p_{+j}. \tag{2.11}$$

Each sum contains the same function, $f(u) = u \log u$. Expanding this function lets us translate probability errors into changes in MI. The derivative needed for its first-order Taylor expansion follows directly from the product rule:

$$f'(u) = \frac{d}{du}\{u \log u\} = \log u + u\frac{1}{u} = \log u + 1. \tag{2.12}$$

The general first-order Taylor formula is

$$f(x) \approx f(a) + f'(a)(x - a), \tag{2.13}$$

where a is the value around which the function is expanded and $x-a$ is the distance from that value. Here the new value is the estimated probability $x = \widehat{u}$, the expansion point is the population probability $a = u$, and $f(t) = t \log t$. Therefore, $x - a = \widehat{u} - u$, which is the sampling error in the estimated probability. Substituting these quantities into the general formula gives

$$\widehat{u} \log \widehat{u} \approx \underbrace{u \log u}_{\text{population value}} + \underbrace{(\log u + 1)(\widehat{u} - u)}_{\text{first-order change}}. \tag{2.14}$$

Subtracting the population value from both sides makes the change explicit:

$$\widehat{u} \log \widehat{u} - u \log u \approx (\log u + 1)(\widehat{u} - u). \tag{2.15}$$

Applying this expansion to the three sums in Equation (2.11) gives

$$\begin{aligned} \widehat{I}(P) - I(P) \approx & \sum_{i,j} (\log p_{ij} + 1)(\widehat{p}_{ij} - p_{ij}) \\ & - \sum_{i} (\log p_{i+} + 1)(\widehat{p}_{i+} - p_{i+}) \\ & - \sum_{j} (\log p_{+j} + 1)(\widehat{p}_{+j} - p_{+j}). \end{aligned} \tag{2.16}$$

The margin errors are sums of the corresponding cell errors:

$$\widehat{p}_{i+} - p_{i+} = \sum_{j} (\widehat{p}_{ij} - p_{ij}), \qquad \widehat{p}_{+j} - p_{+j} = \sum_{i} (\widehat{p}_{ij} - p_{ij}). \tag{2.17}$$

For example, the row-margin sum can now be written as

$$\begin{aligned} \sum_{i} (\log p_{i+} + 1)(\widehat{p}_{i+} - p_{i+}) &= \sum_{i} \left[(\log p_{i+} + 1) \left\{ \sum_{j} (\widehat{p}_{ij} - p_{ij}) \right\} \right] \\ &= \sum_{i,j} (\log p_{i+} + 1)(\widehat{p}_{ij} - p_{ij}). \end{aligned} \tag{2.18}$$

The inner sum over j is the probability error for a fixed row i. It is multiplied by that row's coefficient $\log p_{i+} + 1$; the outer sum then adds these products over all rows. The column-margin sum can be rearranged in the same way.

After this rearrangement, the same cell error $\widehat{p}_{ij} - p_{ij}$ appears in the joint term, its row-margin term and its column-margin term. Its combined coefficient is

$$\begin{aligned}
&(\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \\
&\quad = \log\left(\frac{p_{ij}}{p_{i+}p_{+j}}\right) - 1 \\
&\quad = \ell_P(i,j) - 1. \qquad (2.19)
\end{aligned}$$

Therefore,

$$\widehat{I}(P) - I(P) \approx \sum_{i,j}\{\ell_P(i,j) - 1\}(\widehat{p}_{ij} - p_{ij}). \qquad (2.20)$$

The empirical and population cell probabilities both sum to one, so

$$\sum_{i,j}(\widehat{p}_{ij} - p_{ij}) = \sum_{i,j}\widehat{p}_{ij} - \sum_{i,j}p_{ij} = 1 - 1 = 0. \qquad (2.21)$$

The constant -1 in Equation (2.19) therefore contributes zero. Explicitly,

$$\begin{aligned}
\widehat{I}(P) - I(P) &\approx \sum_{i,j}\{\ell_P(i,j) - 1\}(\widehat{p}_{ij} - p_{ij}) \\
&= \sum_{i,j}\ell_P(i,j)(\widehat{p}_{ij} - p_{ij}) - \sum_{i,j}(\widehat{p}_{ij} - p_{ij}) \\
&= \sum_{i,j}\ell_P(i,j)(\widehat{p}_{ij} - p_{ij}). \qquad (2.22)
\end{aligned}$$

The complicated changes in the joint table and its two margins have therefore reduced to a weighted sum of cell-probability errors. The weight for a cell is its population pointwise MI.

Step 2: From cell errors to observation-level errors.

Let $Z_a = (X_a, Y_a)$ be observation a. An empirical cell probability is the proportion of observations in that cell, so

$$\widehat{p}_{ij} - p_{ij} = \frac{1}{n}\sum_{a=1}^{n}\left\{\mathbf{1}(Z_a = (i,j)) - p_{ij}\right\}. \qquad (2.23)$$

Substituting this expression into Equation (2.22) first gives

$$\widehat{I}(P) - I(P) \approx \sum_{i,j}\ell_P(i,j)\left[\frac{1}{n}\sum_{a=1}^{n}\{\mathbf{1}(Z_a = (i,j)) - p_{ij}\}\right]. \qquad (2.24)$$

The finite sums can be exchanged, so the contribution from observation a is gathered inside one pair of brackets:

$$\begin{aligned}\widehat{I}(P) - I(P) &\approx \frac{1}{n}\sum_{a=1}^{n}\left[\sum_{i,j}\ell_P(i,j)\mathbf{1}(Z_a=(i,j)) - \sum_{i,j}p_{ij}\ell_P(i,j)\right] \\ &= \frac{1}{n}\sum_{a=1}^{n}\{\ell_P(Z_a) - I(P)\}\,. \end{aligned} \tag{2.25}$$

The first sum inside the brackets selects the single cell containing Z_a, giving $\ell_P(Z_a)$. The second is the probability-weighted average of all pointwise-MI values, which is $I(P)$.

To first order, Equation (2.25) expresses the MI error as an average of n observation-level contributions. The contribution of an observation in cell (i,j) is simply that cell's pointwise MI minus the population average, $\ell_P(i,j) - I(P)$. Its mean is zero because

$$\begin{aligned}\mathrm{E}_P\{\ell_P(X,Y) - I(P)\} &= \sum_{i,j}p_{ij}\{\ell_P(i,j) - I(P)\} \\ &= \underbrace{\sum_{i,j}p_{ij}\ell_P(i,j)}_{I(P)} - I(P)\underbrace{\sum_{i,j}p_{ij}}_{1} = 0. \end{aligned} \tag{2.26}$$

The centred function $\ell_P(i,j) - I(P)$ is called the influence function of MI. Equation (2.25) uses it to approximate the sampling error (Hampel, 1974; van der Vaart, 1998; Kandasamy et al., 2015).

Step 3: Calculate the sampling variance.

Define the population variance of these centred contributions by

$$V(P) = \mathrm{Var}_P\{\ell_P(X,Y) - I(P)\} = \sum_{i,j}p_{ij}\{\ell_P(i,j) - I(P)\}^2. \tag{2.27}$$

Under independent sampling, the n terms in Equation (2.25) are independent and each has variance $V(P)$. The covariance between any two different observations is therefore zero. Hence

$$\begin{aligned}\mathrm{Var}\{\widehat{I}(P)\} &\approx \mathrm{Var}\left\{\frac{1}{n}\sum_{a=1}^{n}\{\ell_P(Z_a) - I(P)\}\right\} \\ &= \frac{1}{n^2}\sum_{a=1}^{n}\mathrm{Var}\{\ell_P(Z_a) - I(P)\} \\ &= \frac{1}{n^2}\sum_{a=1}^{n}V(P) = \frac{nV(P)}{n^2} = \frac{V(P)}{n}. \end{aligned} \tag{2.28}$$

Averaging n independent contributions therefore divides their variance by n. For dependent observations, covariance terms must also be included in Equation (2.28), as studied by Moddemeijer (1999). The present experiment uses independent observations.

When $V(P) > 0$, the central limit theorem also gives

$$\frac{\widehat{I}(P) - I(P)}{\sqrt{V(P)/n}} \xrightarrow{d} N(0,1). \tag{2.29}$$

This gives a large-sample normal approximation for a fixed table with positive population cell probabilities. The $1/n$ leading bias from Section 2.2 decreases faster than the $1/\sqrt{n}$ sampling error, so it does not change this limiting distribution.

The independence boundary.

At exact independence, $p_{ij} = p_{i+}p_{+j}$, so $\ell_P(i,j) = \log 1 = 0$ in every cell. Hence $I(P) = 0$, $V(P) = 0$, and the first-order average in Equation (2.25) vanishes. A finite sample can still have positive MI. Its leading variation comes from the squared cell-probability errors in a second-order Taylor expansion, which produces the usual chi-squared limiting distribution (Brillinger, 2004; Marinescu and Balcau, 2025). Testing independence therefore uses a chi-squared distribution rather than the normal approximation for comparing positive MI values.

2.4 Approaches to testing MI differences

This section reviews methods for estimating uncertainty in MI and testing differences between two MI values. The question in this thesis is whether two populations have the same MI, $I(P) = I(Q)$, even when their cell probabilities differ.

The Normal Wald test.

A Wald test estimates a parameter difference, divides it by an estimated standard error, and compares the result with a standard normal distribution (Wald, 1943; van der Vaart, 1998). For MI estimated from independent samples from P and Q, the first-order variance contributions add:

$$\operatorname{Var}\{\widehat{I}(P) - \widehat{I}(Q)\} \approx \frac{V(P)}{n_P} + \frac{V(Q)}{n_Q}. \tag{2.30}$$

Each term is the sampling variance of one MI estimate, and independence of the two samples makes their covariance zero. Replacing the population variances by

estimates gives the standardisation

$$\frac{\widehat{I}(P) - \widehat{I}(Q)}{\sqrt{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q}} \xrightarrow{d} N(0,1) \qquad \text{under } I(P) = I(Q). \tag{2.31}$$

Here $\widehat{V}(P)$ and $\widehat{V}(Q)$ estimate the pointwise-MI variances $V(P)$ and $V(Q)$. This limit assumes positive population cell probabilities, positive pointwise-MI variances and increasing sample sizes. Chapter 3 applies the leading bias correction to the numerator and specifies the full Normal Wald test. With smaller samples, the MI difference may not follow the normal approximation closely, and the estimated standard error can vary between samples. The Wald test still uses the same standard normal distribution cutoff, regardless of how much the estimated standard error varies.

The Welch–Satterthwaite principle.

This thesis adapts the Welch–Satterthwaite principle from comparing population means to comparing population MI values. Welch's test allows the population variances to differ and compares the standardised difference between two sample means with a Student t distribution with ν degrees of freedom (Welch, 1947). For two independent samples with sample means $\bar{X}_P, \bar{X}_Q$ and sample variances s_P^2, s_Q^2, the statistic for testing equal population means and the effective degrees of freedom are

$$T_{\text{Welch}} = \frac{\bar{X}_P - \bar{X}_Q}{\sqrt{s_P^2/n_P + s_Q^2/n_Q}}, \tag{2.32}$$

$$\nu = \frac{\left(s_P^2/n_P + s_Q^2/n_Q\right)^2}{\dfrac{(s_P^2/n_P)^2}{n_P - 1} + \dfrac{(s_Q^2/n_Q)^2}{n_Q - 1}}. \tag{2.33}$$

The denominator of the statistic contains one estimated variance contribution from each sample. Under normal sampling, their component degrees of freedom are $n_P - 1$ and $n_Q - 1$. If s_P^2/n_P dominates, the formula makes the combined ν close to $n_P - 1$; when both contributions matter, both samples' degrees of freedom affect ν. Satterthwaite obtains this form by matching the mean and variance of their sum to a scaled chi-squared variable (Satterthwaite, 1946).

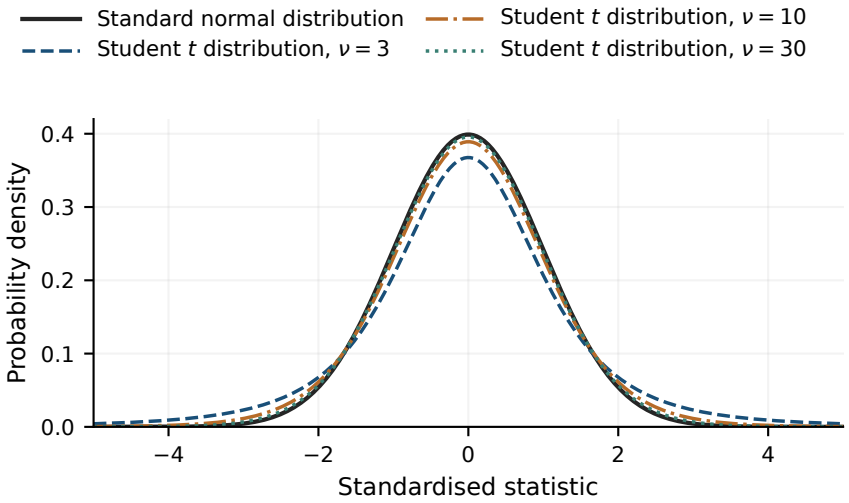


Figure 2.1: Standard normal distribution and Student t distributions with 3, 10 and 30 degrees of freedom.

Figure 2.1 shows that the Student t distribution has heavier tails than the standard normal distribution, especially when ν is small. More probability in the tails means a larger absolute statistic is needed to reject the null at the same significance level. As ν increases, the Student t distribution approaches the standard normal distribution, and their rejection cutoffs become closer.

For MI, the estimated variance contributions are $\widehat{V}(P)/n_P$ and $\widehat{V}(Q)/n_Q$. The ordinary $n_P - 1$ and $n_Q - 1$ values follow from the scaled chi-squared distribution of a sample variance under normal sampling; they do not automatically apply to an estimated MI variance. Chapter 4 therefore derives the component degrees of freedom from the sampling variability of $\widehat{V}(P)$ and $\widehat{V}(Q)$, using the same moment-matching principle.

This distinction matters because the size of a variance estimate and its stability answer different questions. The estimate $\widehat{V}(P)/n_P$ describes the expected spread of an MI estimate across samples. The variability of $\widehat{V}(P)$ describes how much that assessment of spread itself changes across samples. Two variance estimators with the same mean can have different variability. Satterthwaite's calculation uses both moments: greater variability relative to the squared mean gives fewer degrees of freedom and a larger rejection cutoff (Satterthwaite, 1946). A sample-size assignment alone does not adapt to differences in this relative variability between probability tables.

The same principle has been used in regression to account for uncertainty in estimated variances (Pan and Wall, 2002; Bell and McCaffrey, 2002; Kauermann and Carroll, 2001; Fay and Graubard, 2001; Imbens and Kolesár, 2016). For example,

Pan and Wall (2002) calculate approximate degrees of freedom by examining how a regression variance estimate varies between samples, and Expanded Welch uses the variability of the estimated MI variance to choose degrees of freedom in the same way. These regression precedents address variance estimates that are too small or confidence intervals that cover too rarely (Bell and McCaffrey, 2002; Kauermann and Carroll, 2001). A larger cutoff from the Student t distribution can widen the intervals and reduce excessive rejection in such settings.

Previous work on MI uncertainty and comparisons.

Roulston (1999) derives entropy and MI error estimates using error propagation and examines them numerically. These calculations describe the uncertainty of an individual estimate, a necessary input to a comparison of two estimates. The first-order results of Moddemeijer (1989) and Brillinger (2004) give this uncertainty as $V(P)/n_P$, with $V(P)$ determined by the population table. Replacing its probabilities by observed proportions produces the variance estimate used by Wald. This supplies a standard error for the MI difference. Choosing a Welch–Satterthwaite cutoff requires one further quantity: how much the estimated variance varies when a new sample produces a different table.

A Bayesian approach describes uncertainty about population MI given an observed table. Hutter (2002) and Hutter and Zaffalon (2005) model the unknown cell probabilities to calculate the variance and shape of the resulting MI distribution. Our repeated-sampling comparison instead asks how often Normal Wald and Expanded Welch reject equal MI when new samples are drawn from fixed populations P and Q.

Hutcheson's test is a direct information-theoretic Welch predecessor. It compares two Shannon diversity estimates with an estimated variance and a Student t distribution, using

$$\nu_H = \frac{(s^2_{H,P} + s^2_{H,Q})^2}{(s^2_{H,P})^2/n_P + (s^2_{H,Q})^2/n_Q} \tag{2.34}$$

as its effective degrees of freedom (Hutcheson, 1970). Here each of $s^2_{H,P}$ and $s^2_{H,Q}$ is the estimated sampling variance of the corresponding entropy estimator $\widehat{H}$, and each variance component is assigned its sample size, not $n_i - 1$. Hutcheson's rule assigns degrees of freedom from sample sizes. Expanded Welch instead calculates how the estimated MI variance changes with the table, including the effects of a cell change on its row and column totals.

The entropy comparison establishes that a Student t distribution can be used to compare estimated information quantities, but MI requires more than substituting a different numerator. MI combines the row, column and joint entropies of the

same observations. Their estimation errors are connected, because one cell count contributes to all three. The MI variance formulas in Section 2.3 account for that connection. Expanded Welch also tracks how the estimated MI variance changes: a cell change alters both the weights and the pointwise-MI values whose weighted variance is being estimated. Thus Hutcheson supplies a close comparison framework, while the cell and margin derivatives supply the MI-specific inputs.

In a segregation application, Mora and Ruiz-Castillo (2009) derive a normal approximation for comparing positive population MI between populations and time periods, providing a direct precedent for the Normal Wald baseline. Their later peer-reviewed account places the MI index within the segregation literature (Mora and Ruiz-Castillo, 2011).

By contrast, Guha and Chothia (2014) test whether two continuous distributions are identical using MI between pooled observations and their sample labels. For discrete data, Drake and Guha (2014) develop an MI-based k-sample test for equality of distributions. These tests compare complete distributions, not the MI of two variables within each population.

Resampling alternatives.

A bootstrap approximates sampling variation by repeatedly drawing samples with replacement from the observed data (Efron, 1979). For this comparison, samples would be drawn separately from the two observed tables, using their original sample sizes, and the MI difference recalculated for each pair. Resampling from separate empirical probability tables allows their cell probabilities to differ and provides an estimated sampling distribution of the MI difference. The bootstrap variation around the observed estimate could then be used to construct an uncertainty interval for the population difference and assess whether that interval includes zero.

An ordinary empirical bootstrap cannot sample an unobserved cell, because it assigns that cell zero probability. It can therefore miss error arising from unobserved cells. Paninski (2003) gives a broader warning: confidence intervals based on resampling can be too narrow when information estimates remain biased.

A simple permutation test instead combines the two samples and shuffles their group labels. This gives an exact test when the populations have the same complete distribution, $P = Q$. Our hypothesis requires only $I(P) = I(Q)$: the tables can differ while their MI values are equal. Equal MI alone does not guarantee that this label-permutation test has the stated false-positive rate (Chung and Romano, 2013). Specialised permutation methods allow distinct populations in some parameter comparisons (Janssen, 1997; Chung and Romano, 2013); their validity for equal MI would need to be established.

2.5 Previous work and this thesis's contribution

Existing work provides the leading bias and sampling variance of MI (Miller, 1955; Moddemeijer, 1989; Brillinger, 2004; Paninski, 2003). Hutcheson's entropy comparison assigns degrees of freedom from sample sizes (Hutcheson, 1970), while Pan and Wall (2002) use the variability of a regression variance estimate to choose degrees of freedom. Applying this approach to MI requires calculating how the estimated MI variance changes with the observed table, including changes to the cell probabilities, row and column totals, and pointwise MI values.

This thesis derives how changes in the observed count table affect the estimated MI variance and uses that result to calculate MI-specific degrees of freedom for Expanded Welch. The resulting test retains Normal Wald's MI difference and standard error but compares the statistic with a Student t distribution rather than a standard normal distribution.

The experiment measures errors in the shared MI difference and standard error alongside false-positive rates and power. Paninski (2003) shows that finite-sample information estimates can retain substantial bias after a leading correction. Such bias or an inaccurate first-order variance approximation can undermine both tests, even when Expanded Welch lowers rejection.

Chapter 3

The Two-Sample MI Test and Wald Baseline

Normal Wald and Expanded Welch share an estimated MI difference and standard error. This chapter defines their statistic and the Normal Wald p-value.

3.1 Problem setup and bias-corrected estimator

Let

$$N^P = (N_{ij}^P), \qquad N^Q = (N_{ij}^Q) \tag{3.1}$$

be independent $r \times c$ count tables with totals n_P and n_Q. The rows and columns represent the same categorical variables X and Y in both populations, with the same category definitions. The empirical probabilities are

$$\widehat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \qquad \widehat{q}_{ij} = \frac{N_{ij}^Q}{n_Q}. \tag{3.2}$$

The population parameter is the signed difference

$$\Delta = I(P) - I(Q). \tag{3.3}$$

The principal test is two-sided:

$$H_0 : \Delta = 0 \quad \text{against} \quad H_1 : \Delta \neq 0. \tag{3.4}$$

The large-sample argument uses the following assumptions.

Assumption 1 (Sampling)**.** *Observations are independent and identically distributed within each population, and the two population samples are independent.*

Assumption 2 (Aligned fixed alphabets)**.** *The row and column categories are fixed in advance and aligned between the two tables.*

Assumption 3 (Positive cell probabilities)**.** *Every cell in the population table has a probability greater than zero. A sample can still contain no observations in a cell, especially when that cell is rare. Such samples remain in the study; we record how often cells have zero observed counts to show how sparse the samples are.*

Assumption 4 (Positive MI)**.** *Both populations have positive MI: $I(P) > 0$ and $I(Q) > 0$. With the positive cell probabilities in Assumption 3, this is equivalent to requiring positive first-order variances $V(P) > 0$ and $V(Q) > 0$. It excludes exact independence, where the first-order approximation breaks down.*

Assumption 5 (Logarithm units)**.** *All logarithms in this thesis are natural, so MI differences and standard errors are reported in nats. Changing the logarithm base scales both by the same positive constant and leaves the standardised statistic unchanged. This is a choice of units, not a condition for the test to be valid.*

Bias-corrected estimator.

For the first table, define empirical pointwise MI by

$$\widehat{\ell}_P(i,j) = \log\left(\frac{\widehat{p}_{ij}}{\widehat{p}_{i+}\widehat{p}_{+j}}\right) \qquad \text{when } \widehat{p}_{ij} > 0. \tag{3.5}$$

The plug-in MI estimate is

$$\widehat{I}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij}\widehat{\ell}_P(i,j). \tag{3.6}$$

The same calculation gives $\widehat{I}(Q)$.

The leading fixed-support MI bias appears in Equation (2.8) (Miller, 1955; Moddemeijer, 1989). Subtracting it from each plug-in estimate gives the estimated effect:

$$\widehat{\Delta}_{\mathrm{BC}} = \left\{\widehat{I}(P) - \frac{(r-1)(c-1)}{2n_P}\right\} - \left\{\widehat{I}(Q) - \frac{(r-1)(c-1)}{2n_Q}\right\}. \tag{3.7}$$

The two terms in braces are the bias-corrected MI estimates for the separate populations. The correction uses the declared row and column categories because every population cell has positive probability under the model, even when some cells are unobserved in a sample. Corrected estimates can be negative; those values are retained when calculating the difference. When $n_P = n_Q$, the two corrections cancel, so $\widehat{\Delta}_{\mathrm{BC}}$ is exactly the raw plug-in MI difference. With unequal sample sizes,

their difference removes the leading unequal-sample bias described in Section 2.2. A correction that estimates support from occupied cells would be a different estimator.

3.2 Shared standard error and test statistic

Section 2.3 showed that the first-order sampling variance of $\widehat{I}(P)$ is $V(P)/n_P$, where $V(P)$ is the population variance of pointwise MI (Moddemeijer, 1989; Brillinger, 2004). Subtracting the fixed leading bias term in Equation (3.7) does not change this variance.

The observed table estimates $V(P)$ by

$$\widehat{V}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij}\{\widehat{\ell}_P(i,j) - \widehat{I}(P)\}^2. \tag{3.8}$$

Define $\widehat{V}(Q)$ analogously.

Because the two samples are independent, their estimated variance contributions add:

$$\widehat{\mathrm{SE}}^2 = \frac{\widehat{V}(P)}{n_P} + \frac{\widehat{V}(Q)}{n_Q}. \tag{3.9}$$

The common standardised statistic is

$$T = \frac{\widehat{\Delta}_{\mathrm{BC}}}{\widehat{\mathrm{SE}}}. \tag{3.10}$$

The sign of T indicates whether P or Q has the larger corrected MI estimate, while its magnitude measures the difference relative to its estimated standard error. Under H_0, we compare T with a reference distribution centred at zero. Equations (3.7)–(3.10) are shared by Normal Wald and Expanded Welch. Their difference lies in the reference distribution, not the effect estimate or plug-in standard error.

3.3 Normal Wald test and motivation for Expanded Welch

When both sample sizes grow while the table dimensions and positive population probabilities stay fixed, the statistic approaches a standard normal distribution under equal population MI (Wald, 1943; van der Vaart, 1998):

$$T \xrightarrow{d} N(0,1) \qquad \text{under } H_0. \tag{3.11}$$

The two-sided Normal Wald p-value is

$$p_{\text{Wald}} = 2\{1 - \Phi(|T|)\}, \tag{3.12}$$

where Φ is the standard normal distribution function.

Why use a Student t distribution?

Wald uses the standard normal distribution at every sample size, even though its standard error is estimated from the observed tables. Expanded Welch uses the variability of that variance estimate to choose a Student t distribution instead. Hutcheson's entropy test and ordinary Welch's test set each variance component's degrees of freedom from its sample size: n_i (Hutcheson, 1970) or $n_i - 1$ (Welch, 1947). Neither assignment measures how cell probabilities and margins make the estimated MI variance change between samples.

To choose the degrees of freedom, we need the average value of each estimated MI variance and how much that estimate varies across samples (Satterthwaite, 1946). For P, to first order,

$$\mathrm{E}\{\widehat{V}(P)\} \approx V(P). \tag{3.13}$$

Chapter 4 derives $\mathrm{Var}\{\widehat{V}(P)\}$ by tracking how the calculated MI variance changes when one observation changes the joint table.

Chapter 4

The Expanded Welch–Satterthwaite Test for MI

Expanded Welch keeps the estimated MI difference, standard error and test statistic from Chapter 3, but changes the distribution used to judge that statistic from the standard normal distribution to a Student t distribution. To choose its degrees of freedom, this chapter calculates how much the estimated MI variance changes between samples. The name Expanded Welch refers to this expansion of the MI variance as the table probabilities change.

4.1 Choosing degrees of freedom for Expanded Welch

The Welch–Satterthwaite equation combines the two estimated MI variance contributions (Satterthwaite, 1946; Welch, 1947):

$$\nu = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\left\{\widehat{V}(P)/n_P\right\}^2/\nu_P + \left\{\widehat{V}(Q)/n_Q\right\}^2/\nu_Q}. \tag{4.1}$$

The terms $\widehat{V}(P)/n_P$ and $\widehat{V}(Q)/n_Q$ are the estimated variance contributions used to calculate the standard error for both Normal Wald and Expanded Welch in Chapter 3. The new variables being introduced are the component degrees of freedom ν_P and ν_Q. They describe how much each estimated MI variance varies between samples relative to its mean.

The derivation has four steps. First, we relate the degrees of freedom to the average and across-sample variation of $\widehat{V}(P)$. Second, we calculate how $V(P)$

changes when a small amount of probability moves into one cell. Third, that change tells us how variable $\widehat{V}(P)$ is. Finally, we combine the degrees of freedom from P and Q using Equation (4.1).

Moment matching for one MI variance estimate.

Satterthwaite approximates the nonnegative variance estimate $\widehat{V}(P)$ by the scaled chi-squared variable

$$\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_P}}{\nu_P}. \tag{4.2}$$

The scale and degrees of freedom are chosen to match the mean and variance of $\widehat{V}(P)$.

For a chi-squared variable with ν_P degrees of freedom,

$$\mathrm{E}(\chi^2_{\nu_P}) = \nu_P, \qquad \mathrm{Var}(\chi^2_{\nu_P}) = 2\nu_P. \tag{4.3}$$

The scaling in Equation (4.2) therefore matches the mean automatically:

$$\mathrm{E}\left[\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_P}}{\nu_P}\right] = \mathrm{E}\{\widehat{V}(P)\}. \tag{4.4}$$

Its variance is

$$\mathrm{Var}\left[\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_P}}{\nu_P}\right] = \frac{[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu_P^2}\,\mathrm{Var}(\chi^2_{\nu_P}) \tag{4.5}$$

$$= \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu_P}. \tag{4.6}$$

Matching this quantity to $\mathrm{Var}\{\widehat{V}(P)\}$ gives

$$\boxed{\nu_P = \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\mathrm{Var}\{\widehat{V}(P)\}}}. \tag{4.7}$$

The numerator depends on the mean of the estimated MI variance; the denominator measures how much that estimate fluctuates between samples. To first order, $\mathrm{E}\{\widehat{V}(P)\} \approx V(P)$, so the remaining task is to calculate $\mathrm{Var}\{\widehat{V}(P)\}$.

4.2 How the MI variance changes with the table

To find how much the estimated MI variance changes between samples, we first ask how the population variance responds when a small amount of probability is moved into one cell (Hampel, 1974; van der Vaart, 1998). Fix a cell (x, y) and define the perturbed distribution

$$P_\varepsilon = (1-\varepsilon)P + \varepsilon\delta_{(x,y)}, \tag{4.8}$$

where $\delta_{(x,y)}$ places probability one on cell (x, y). At $\varepsilon = 0$, the distribution is P. Increasing ε moves a small amount of probability toward the selected cell while preserving total probability one.

For an arbitrary cell (i, j),

$$p_{ij}(\varepsilon) = (1-\varepsilon)p_{ij} + \varepsilon\mathbf{1}\{i = x, j = y\}. \tag{4.9}$$

The joint and marginal probability derivatives at the original distribution are

$$\left.\frac{\mathrm{d}p_{ij}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{i = x, j = y\} - p_{ij}, \tag{4.10}$$

$$\left.\frac{\mathrm{d}p_{i+}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{i = x\} - p_{i+}, \tag{4.11}$$

$$\left.\frac{\mathrm{d}p_{+j}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{j = y\} - p_{+j}. \tag{4.12}$$

The derivatives of the joint probabilities sum to zero because

$$\sum_{i,j}\{\mathbf{1}(i = x, j = y) - p_{ij}\} = 1 - 1 = 0. \tag{4.13}$$

Effect on pointwise MI and MI.

Along the perturbation path, pointwise MI is

$$\ell_{P_\varepsilon}(i, j) = \log p_{ij}(\varepsilon) - \log p_{i+}(\varepsilon) - \log p_{+j}(\varepsilon). \tag{4.14}$$

For each logarithm, the derivative is the probability derivative divided by the probability. Substituting the joint and marginal derivatives gives

$$\begin{aligned}\left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\ell_{P_\varepsilon}(i,j)\right|_0 &= \frac{\mathbf{1}\{i=x,j=y\}-p_{ij}}{p_{ij}} - \frac{\mathbf{1}\{i=x\}-p_{i+}}{p_{i+}} - \frac{\mathbf{1}\{j=y\}-p_{+j}}{p_{+j}} \\ &= \frac{\mathbf{1}\{i=x,j=y\}}{p_{ij}} - \frac{\mathbf{1}\{i=x\}}{p_{i+}} - \frac{\mathbf{1}\{j=y\}}{p_{+j}} + 1. \end{aligned} \tag{4.15}$$

The joint term supplies -1, while subtracting the row and column terms supplies $+1$ each. Their sum is the final $+1$.

MI along the path is

$$I(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon)\ell_{P_\varepsilon}(i,j). \tag{4.16}$$

The product rule separates its derivative into the change in probability weights and the change in pointwise-MI values:

$$\begin{aligned}\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 &= \sum_{i,j}\{\mathbf{1}(i=x,j=y)-p_{ij}\}\ell_P(i,j) \\ &\quad + \sum_{i,j} p_{ij}\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0 . \end{aligned} \tag{4.17}$$

The first sum is $\ell_P(x,y) - I(P)$. In the second sum, the selected-cell term contributes $p_{xy}/p_{xy} = 1$. The row term contributes $-\sum_j p_{xj}/p_{x+} = -1$; the column term likewise contributes -1; and the constant contributes $\sum_{i,j} p_{ij} = 1$. These four contributions cancel. Hence

$$\boxed{\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \ell_P(x,y) - I(P)}. \tag{4.18}$$

This is the centred pointwise-MI influence function from Section 2.3.

Effect on the MI variance.

The target is the first-order change in

$$V(P) = \mathrm{Var}_P\{\ell_P(X,Y)\}. \tag{4.19}$$

Write this variance in second-moment form:

$$V(P) = \mathrm{E}_P\{\ell_P(X,Y)^2\} - I(P)^2. \tag{4.20}$$

Define the observation-level variance sensitivity by

$$g_P(x,y) = \left.\frac{\mathrm{d}V(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0. \tag{4.21}$$

Differentiating Equation (4.20) gives

$$g_P(x,y) = \left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\,\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\}\right|_0 - 2I(P)\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0. \tag{4.22}$$

It remains to differentiate

$$\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\} = \sum_{i,j} p_{ij}(\varepsilon)\ell_{P_\varepsilon}(i,j)^2. \tag{4.23}$$

The product rule gives

$$\begin{aligned}\left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\,\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\}\right|_0 &= \sum_{i,j}\left.\frac{\mathrm{d}p_{ij}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 \ell_P(i,j)^2 \\ &\quad + 2\sum_{i,j} p_{ij}\ell_P(i,j)\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0.\end{aligned} \tag{4.24}$$

The first sum is

$$\ell_P(x,y)^2 - \mathrm{E}_P\{\ell_P(X,Y)^2\}. \tag{4.25}$$

For the second sum, define the probability-weighted row and column means

$$\mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} = \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}}, \tag{4.26}$$

$$\mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} = \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}}. \tag{4.27}$$

Substitution of Equation (4.15) gives

$$\begin{aligned}&\sum_{i,j} p_{ij}\ell_P(i,j)\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0 \\ &\quad = \ell_P(x,y) - \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}} - \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}} + I(P) \\ &\quad = \ell_P(x,y) - \mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} - \mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P).\end{aligned} \tag{4.28}$$

Combining Equations (4.18), (4.22), (4.25), and (4.28) gives the variance sensitivity. To see how its squared term arises, combine the weight-change term in

Equation (4.25) with the derivative of $-I(P)^2$:

$$\begin{aligned}
&\ell_P(x,y)^2 - \mathrm{E}_P\{\ell_P(X,Y)^2\} - 2I(P)\{\ell_P(x,y) - I(P)\} \\
&\quad = \ell_P(x,y)^2 - \{V(P) + I(P)^2\} - 2I(P)\ell_P(x,y) + 2I(P)^2 \\
&\quad = \ell_P(x,y)^2 - 2I(P)\ell_P(x,y) + I(P)^2 - V(P) \\
&\quad = \{\ell_P(x,y) - I(P)\}^2 - V(P).
\end{aligned}$$

The second line uses $\mathrm{E}_P\{\ell_P(X,Y)^2\} = V(P) + I(P)^2$; the last line completes the square. Adding twice the expression in Equation (4.28) gives

$$\boxed{\begin{aligned}
g_P(x,y) = \{\ell_P(x,y) - I(P)\}^2 - V(P) \\
+ 2\Big[\ell_P(x,y) - \mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} \\
- \mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P)\Big].
\end{aligned}} \tag{4.29}$$

The first line is the change if the pointwise-MI values were held fixed. The bracket accounts for the fact that they change too: each pointwise-MI value uses a cell probability and its row and column totals. The conditional means collect the effects of those changing totals. Both parts are needed to describe how the estimated variance varies across samples.

4.3 Component degrees of freedom and final test

Population component degrees of freedom.

Define the population variability of the observation-level sensitivities by

$$\tau^2(P) = \mathrm{Var}_P\{g_P(X,Y)\}. \tag{4.30}$$

The sensitivities average to zero under P, as shown in Appendix A. Their variation, not their average, determines how much the estimated MI variance changes between samples.

Assumption 6 (Regular variance sensitivity)**.** *For the first-order Expanded Welch construction,* $\tau^2(P) > 0$ *and* $\tau^2(Q) > 0$.

An observation in cell $Z_a^{(P)}$ can be represented by a table that places probability one in that cell, denoted $\delta_{Z_a^{(P)}}$. Relative to P, that observation contributes the table change $\delta_{Z_a^{(P)}} - P$. The empirical table averages these one-observation tables, so its difference from P is

$$\widehat{P} - P = \frac{1}{n_P} \sum_{a=1}^{n_P} \left\{\delta_{Z_a^{(P)}} - P\right\}. \tag{4.31}$$

The plug-in $\widehat{V}(P)$ applies the variance calculation to this empirical table. Equation (4.21) gives its first-order change for one observation's direction. Because a first-order change is linear in the table error (van der Vaart, 1998, Chapter 20), the changes from the observations can be averaged:

$$\widehat{V}(P) - V(P) \approx \frac{1}{n_P} \sum_{a=1}^{n_P} g_P(Z_a^{(P)}). \tag{4.32}$$

The observations are independent, so

$$\mathrm{Var}\{\widehat{V}(P)\} \approx \frac{1}{n_P^2} \sum_{a=1}^{n_P} \mathrm{Var}_P\{g_P(Z_a^{(P)})\} \tag{4.33}$$

$$= \frac{\tau^2(P)}{n_P}. \tag{4.34}$$

Substitution into Equation (4.7) gives the population component degree of freedom

$$\boxed{\nu_P \approx \frac{2n_P V(P)^2}{\tau^2(P)}}. \tag{4.35}$$

To interpret Equation (4.35), divide the sampling variance of $\widehat{V}(P)$ by $V(P)^2$, the square of the population pointwise-MI variance:

$$\frac{\mathrm{Var}\{\widehat{V}(P)\}}{V(P)^2} \approx \frac{\tau^2(P)}{n_P V(P)^2}, \tag{4.36}$$

so the component degrees of freedom are approximately twice the reciprocal of this relative variance. When the estimate fluctuates little compared with the size of $V(P)$, the degrees of freedom are large and the Student t distribution is close to the standard normal distribution, whereas greater relative fluctuation gives fewer degrees of freedom and a larger rejection cutoff. Positive $V(P)$ does not guarantee positive $\tau^2(P)$: table changes can have no first-order effect on $V(P)$. In that case, describing the sampling variation of $\widehat{V}(P)$ requires a higher-order approximation.

Table-based component degrees of freedom.

The observed table replaces every population quantity in Equation (4.29) by its empirical counterpart. The sums below are over occupied cells, because pointwise MI is undefined at an empirical zero. For each occupied cell,

$$\begin{aligned} \widehat{g}_P(i,j) = {} & \{\widehat{\ell}_P(i,j) - \widehat{I}(P)\}^2 - \widehat{V}(P) \\ & + 2\left[\widehat{\ell}_P(i,j) - \frac{\sum_{j'} \widehat{p}_{ij'} \widehat{\ell}_P(i,j')}{\widehat{p}_{i+}} - \frac{\sum_{i'} \widehat{p}_{i'j} \widehat{\ell}_P(i',j)}{\widehat{p}_{+j}} + \widehat{I}(P)\right]. \end{aligned} \tag{4.37}$$

In all weighted sums, an empty cell contributes zero; no logarithm or sensitivity needs to be evaluated there. The empirical variance is centred explicitly:

$$\widehat{\tau}^2(P) = \sum_{i,j} \widehat{p}_{ij} \left\{ \widehat{g}_P(i,j) - \sum_{k,l} \widehat{p}_{kl} \widehat{g}_P(k,l) \right\}^2 . \tag{4.38}$$

The estimated component degree of freedom is

$$\boxed{\widehat{\nu}_P = \frac{2n_P \widehat{V}(P)^2}{\widehat{\tau}^2(P)}}. \tag{4.39}$$

Repeating the same calculation for the second table gives $\widehat{\nu}_Q$.

Combining the two samples' degrees of freedom.

Substituting the two variance contributions and component degrees of freedom into Equation (4.1) gives

$$\boxed{\widehat{\nu}_{\text{expanded}} = \frac{\left\{ \widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q \right\}^2}{\left\{ \widehat{V}(P)/n_P \right\}^2 / \widehat{\nu}_P + \left\{ \widehat{V}(Q)/n_Q \right\}^2 / \widehat{\nu}_Q}}. \tag{4.40}$$

The expanded two-sided p-value is

$$p_{\text{expanded}} = 2\left\{1 - F_t(|T|; \widehat{\nu}_{\text{expanded}})\right\}, \tag{4.41}$$

where $F_t(x;\nu)$ denotes the cumulative distribution function of a Student t distribution with ν degrees of freedom, evaluated at x.

For a fixed finite T, the Student t distribution gives a two-sided p-value at least as large as the standard normal distribution used by Wald. Under the implemented validity rules, every valid Expanded Welch result is also Wald-valid, so every Expanded Welch rejection is also a Wald rejection, including when invalid outputs count as non-rejections in the reported rates. Expanded Welch cannot raise rejection where Wald is already too conservative. Appendix A.5 examines a zero-component case in the combined degrees-of-freedom formula.

4.4 Worked example and implementation

Consider the two observed 3×3 tables

$$N^P = \begin{pmatrix} 35 & 10 & 5 \\ 10 & 20 & 5 \\ 5 & 5 & 10 \end{pmatrix}, \qquad N^Q = \begin{pmatrix} 30 & 5 & 5 \\ 5 & 30 & 5 \\ 5 & 5 & 10 \end{pmatrix}. \tag{4.42}$$

Their totals are $n_P = 105$ and $n_Q = 100$. In N^P, cell $(1,1)$ has count 35, and its row and column each total 50. Its empirical pointwise MI is therefore

$$\widehat{\ell}_P(1,1) = \log\left\{\frac{35/105}{(50/105)(50/105)}\right\} = \log(1.47) \simeq 0.38526.$$

This cell contributes $(35/105)(0.38526)$ to the plug-in MI estimate. Summing the contributions from all nine cells gives $\widehat{I}(P) = 0.13693$; the same calculation for N^Q gives $\widehat{I}(Q) = 0.25848$ nats. Since $(3-1)(3-1) = 4$, the corrected difference is

$$\widehat{\Delta}_{\mathrm{BC}} = \left(0.13693 - \frac{4}{2(105)}\right) - \left(0.25848 - \frac{4}{2(100)}\right) = -0.12059. \tag{4.43}$$

The variance calculation weights each squared deviation from the estimated MI by its cell probability. Cell $(1,1)$, for example, contributes $(35/105)(0.38526 - 0.13693)^2 \simeq 0.02056$ to $\widehat{V}(P)$. Summing over all cells gives $\widehat{V}(P) = 0.27189$ and $\widehat{V}(Q) = 0.43371$. Hence

$$\widehat{\mathrm{SE}} = \left(\frac{0.27189}{105} + \frac{0.43371}{100}\right)^{1/2} = 0.08323, \qquad T = -1.44898. \tag{4.44}$$

Normal Wald gives $p = 0.14734$.

For Expanded Welch, the same cell illustrates the sensitivity calculation. The first row's pointwise-MI values are 0.38526, -0.51083 and -0.64436, weighted by its counts 35, 10 and 5. Its conditional mean is

$$\frac{35(0.38526) + 10(-0.51083) + 5(-0.64436)}{50} \simeq 0.10308.$$

Because this count table is symmetric, the first column has the same conditional mean. Substituting these values into Equation (4.37) gives

$$\begin{aligned}\widehat{g}_P(1,1) &\simeq (0.38526 - 0.13693)^2 - 0.27189 \\ &\quad + 2\{0.38526 - 0.10308 - 0.10308 + 0.13693\} \\ &\simeq 0.4218.\end{aligned}$$

The probability-weighted mean of the sensitivities is zero, so this cell contributes $(35/105)(0.4218)^2 \simeq 0.0593$ to $\widehat{\tau}^2(P)$. Repeating the calculation for all cells in both tables gives $\widehat{\tau}^2(P) = 0.93212$ and $\widehat{\tau}^2(Q) = 0.82035$. Substitution into Equation (4.39) gives component degrees of freedom

$$\widehat{\nu}_P = 16.65, \qquad \widehat{\nu}_Q = 45.86. \tag{4.45}$$

Combining them with Equation (4.40) gives $\widehat{\nu}_{\text{expanded}} = 59.03$ and $p = 0.15264$. Neither method rejects at the 5% level.

Operational algorithm and complexity.

Table 4.1 summarises one comparison.

Table 4.1: Operational Expanded Welch–Satterthwaite algorithm.

Step	Calculation
1	Validate the two aligned count tables and their declared alphabet dimensions; independence of samples is a design assumption.
2	Compute empirical probabilities, pointwise MI, plug-in MI, and the leading bias-corrected difference.
3	Compute $\widehat{V}(P)$, $\widehat{V}(Q)$, the shared standard error, and T.
4	Compute the row and column pointwise-MI means needed in $\widehat{g}_P$ and $\widehat{g}_Q$.
5	Compute $\widehat{\tau}^2(P)$, $\widehat{\tau}^2(Q)$, and the two component degrees of freedom.
6	Combine the components using Equation (4.40).
7	Return the p-value from the Student t distribution, effective degrees of freedom, and validity diagnostics.

Each empirical probability, pointwise-MI value, row mean, column mean, and variance contribution is calculated by a fixed number of table scans. The time complexity is therefore $O(rc)$, and the working memory is also $O(rc)$.

4.5 Behaviour and limitations

Near independence.

Consider a population table that approaches independence while its row and column margins stay fixed. At independence, MI is zero and its first-order change is also zero. The leading changes in MI, the pointwise-MI variance $V(P)$, and the variance sensitivity $\tau^2(P)$ can therefore be compared using a second-order expansion. Appendix A.2 gives the calculation and its fixed-margin assumptions. Close to independence on that path, it gives

$$V(P) \approx 2I(P), \qquad \tau^2(P) \approx 4V(P). \tag{4.46}$$

Substituting these local approximations into the component degrees-of-freedom formula gives

$$\nu_P \approx \frac{2n_P V(P)^2}{\tau^2(P)} \approx \frac{n_P V(P)}{2} \approx n_P I(P). \tag{4.47}$$

This population approximation describes the estimated degrees of freedom when sampling errors are small enough for the first-order calculation to be accurate. Smaller population MI then gives fewer degrees of freedom and a larger cutoff from the Student t distribution.

Very close to independence, sampling noise itself creates apparent MI in the empirical table. The resulting variance estimates need not be close to their population values, so the estimated degrees of freedom need not tend to zero with population MI at a fixed sample size. These local expansions assume fixed table dimensions, positive cell probabilities and the fixed-margin path. Appendix A.4 retains the quadratic sampling terms needed when population MI is very small.

Theoretical status and validity conditions.

Equations (4.18) and (4.29) are exact derivatives for positive population tables and provide first-order approximations to the sampling errors in MI and its estimated variance. The scaled chi-squared model in Equation (4.2) and the Student t distribution for T are finite-sample approximations. The MI difference and standard error come from the same tables. A sample that changes the estimated difference can also change its standard error, and Expanded Welch's larger cutoff does not account for how the estimated MI difference and its standard error move together. Appendix A.3 gives the first-order covariance between each population's MI and variance estimates. As sample sizes grow under regular conditions, the estimated degrees of freedom increase and the Student t distribution approaches the standard normal distribution used by Wald.

Both tests need a finite corrected MI difference and a positive, finite estimated standard error, while Expanded Welch also needs positive, finite component and combined degrees of freedom. A failed requirement gives an invalid calculation, not evidence for the null. Appendix B.4 specifies the numerical criteria; the experiment evaluates validity alongside calibration.

Chapter 5

Experimental Design

The experiment uses the simulation-design framework of Morris et al. (2019) to evaluate whether Expanded Welch's larger rejection cutoff improves performance compared with Normal Wald. It first measures false positives when the two populations have equal MI, then examines the estimation errors behind those results, and finally estimates power when MI differs between the two populations. Bias and variance diagnostics link rejection rates to errors in the estimated MI difference and standard error. Each comparison starts from fixed joint probability tables with known MI values, so the true difference is available when assessing the sampled estimates.

5.1 Questions and controlled comparisons

The first study measures false positives when the two populations have equal MI. It varies sample size, table size and category frequencies to determine when Normal Wald and Expanded Welch reject equal MI too often or too rarely relative to the intended 5% rate. It also records how often each test returns a usable result.

The second study examines how the size of the MI values affects Expanded Welch's estimated degrees of freedom. It changes the MI value in both populations while keeping their MI values equal and holding table size and category frequencies fixed. The study includes small positive MI values to examine how the degrees of freedom behave close to independence.

The third study measures statistical power when MI differs between the two populations. It keeps one population fixed and increases the MI of the other, while holding sample size, table size and category frequencies fixed within each comparison. Comparing Normal Wald and Expanded Welch shows how the larger

rejection cutoff affects the probability of detecting a true MI difference. Table 5.1 summarises these studies and the construction and sample-allocation checks.

Table 5.1: Overview of the three main studies and two supporting checks.

Comparison	Factor varied	Main purpose
Equal MI	Sample size, table size and category frequencies	False positives and usable results.
Population MI	MI value, equal in both populations	Degrees of freedom and near-independence behaviour.
Power	True MI difference	Power alongside null calibration.
Construction check	Copying or four-cell transfer	Sensitivity to a different association pattern.
Sample allocation	Relative sample sizes at fixed total	False positives and power with unequal samples.

5.2 Constructing joint distributions with known MI

The construction changes association while preserving the category frequencies. For one population, draw X with category probabilities $a = (a_1, \ldots, a_k)$. With probability λ, set $Y = X$; otherwise draw Y independently with the same probabilities a. Copying makes matching categories more likely than under independence. Both branches give Y the same marginal probabilities, so changing λ does not change the row or column totals.

An observation in a diagonal cell can arrive through either branch, whereas an off-diagonal observation can arrive only through the independent branch. Combining these probabilities gives

$$p_{ij}(\lambda) = (1 - \lambda)a_i a_j + \lambda a_i \mathbf{1}_{\{i=j\}}, \qquad 0 \leq \lambda < 1, \tag{5.1}$$

where the indicator is one when the two categories match and zero otherwise. At $\lambda = 0$, the variables are independent and MI is zero. Increasing λ increases MI, so the target MI determines the required copying probability. Appendix B gives the numerical search and shows why the margins remain fixed.

Constructing a 2×2 joint distribution with positive MI.

For P, take $a = (0.7, 0.3)$ and a copying probability of approximately 0.20347. Cell $(1, 1)$ requires $X = 1$, which has probability 0.7. Given $X = 1$, copying always gives $Y = 1$, whereas an independent draw gives $Y = 1$ with probability 0.7. Thus

$$p_{11} \simeq 0.7\{0.20347 + (1 - 0.20347)(0.7)\} = 0.53273,$$
$$p_{12} \simeq 0.7(1 - 0.20347)(0.3) = 0.16727.$$

The second equation has no copying term because copying cannot produce $X = 1, Y = 2$. The remaining cells follow the same rule. For Q, use $a = (0.8, 0.2)$ and a copying probability of approximately 0.26091. These choices give $I(P) = 0.02$ and $I(Q) = 0.03$ nats:

$$P \simeq \begin{pmatrix} 0.53273 & 0.16727 \\ 0.16727 & 0.13273 \end{pmatrix}, \qquad Q \simeq \begin{pmatrix} 0.68175 & 0.11825 \\ 0.11825 & 0.08175 \end{pmatrix}. \tag{5.2}$$

Both joint distributions have positive MI, and the true difference used to assess power has magnitude 0.01 nats.

Table sizes and margin regimes.

The copying construction is used for square tables of sizes $2 \times 2, 4 \times 4, 8 \times 8$ and 16×16, containing 4, 16, 64 and 256 cells. Each doubling of the number of categories multiplies the number of cells by four, so at a fixed sample size the observations are distributed over more possible cells.

Three margin regimes vary the category frequencies: uniform frequencies in both populations, the same uneven frequencies in both, or different uneven frequencies between them. For uneven margins, the first category has probability ρ, and the remaining mass is shared equally:

$$m_k(\rho) = \left(\rho, \frac{1 - \rho}{k - 1}, \ldots, \frac{1 - \rho}{k - 1}\right). \tag{5.3}$$

Each population uses the same vector for its row and column margins. Table 5.2 specifies the vectors for P and Q. The copying probabilities for P and Q are chosen separately to achieve the targets $I(P)$ and $I(Q)$.

Under equal MI, the uniform and same-skew regimes give identical joint probability tables, $P = Q$. In contrast, the different-skew regime gives different tables with equal MI because the category frequencies differ between populations. Comparing the regimes therefore tests false-positive behaviour under both identical and distinct population tables.

Table 5.2: Row and column margins for each population.

Regime	P: rows and columns	Q: rows and columns
Uniform	$(1/k, \ldots, 1/k)$	$(1/k, \ldots, 1/k)$
Same skew	$m_k(0.8)$	$m_k(0.8)$
Different skew	$m_k(0.7)$	$m_k(0.8)$

Increasing table size under the uneven-margin rule spreads the remaining mass over more categories, changing both the number of cells and their probabilities. This comparison shows what happens under a common margin rule as the table grows. Uniform versus same skew instead compares even and uneven category frequencies at a fixed size; different skew allows the two populations' frequencies to differ as well.

5.3 The three studies and supporting checks

5.3.1 Sample size, table size and margins

The first study fixes $I(P) = I(Q) = 0.02$ nats and crosses all four sizes with all three margin regimes. For each of the twelve fixed population pairs, sample size varies over

$$n_P = n_Q \in \{10, 20, 50, 100, 250, 500, 1000, 2500, 10000, 50000\}. \tag{5.4}$$

These 120 settings show whether the false-positive rate approaches 5% as more observations become available, and whether table size or margins change that progression. Valid-result rates distinguish a genuinely low false-positive rate from a failure to calculate the test.

The common target of 0.02 nats gives positive MI in all four table sizes and three margin regimes. Fixing population MI does not fix the difficulty of estimating it: tables with equal MI can have different cell probabilities and different MI sampling variances. Keeping each population pair fixed throughout the sweep isolates the effect of sample size on test performance. The sweep includes small samples, intermediate samples and large-sample checks so that both calculation failures and convergence are visible.

5.3.2 Population MI and degrees of freedom

The second study keeps 4×4 tables and different-skew margins, and varies the positive MI value while keeping it equal in both populations:

$$I(P) = I(Q) \in \{0.0001, 0.0003, 0.001, 0.003, 0.01, 0.02, 0.05\}. \tag{5.5}$$

The sweep is repeated at $n = 100$ and 1,000. Small positive MI targets approach independence, where the first-order MI variance becomes small, while larger targets show how the tests behave away from that region. Tracking the estimated degrees of freedom connects changes in rejection to the cutoff from the Student t distribution. The two sample sizes show whether these patterns persist with more observations. All targets are positive, so the sweep approaches rather than reaches exact independence.

5.3.3 Power to detect MI differences

The third study fixes $I(P) = 0.02$ and sets $I(Q) = 0.02 + |\Delta|$, using

$$|\Delta| \in \{0, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2\} \quad \text{nats.} \tag{5.6}$$

At $n = 1{,}000$ and 10,000, compare all four sizes under different skew, then all three margin regimes at 4×4. The signed parameter is $\Delta = I(P) - I(Q)$, which is negative when Q has the larger MI; figures use its known nonnegative magnitude.

Within each curve, the entire population table P stays fixed, while Q's copying probability increases to give the specified MI differences. Small differences show where the methods may disagree; larger differences show whether power eventually becomes high. Each curve includes the corresponding equal-MI setting from the null study, so power can be read alongside the false-positive rate at the same sample size.

5.3.4 A supporting change in construction

An eight-setting check asks whether representative findings persist when association is concentrated in four cells. Start from the independence table, add the same amount to cells $(1, 1)$ and $(2, 2)$, and subtract it from $(1, 2)$ and $(2, 1)$. Each affected row and column gains and loses the same amount. This transfer follows the sparse construction of Berrett and Samworth (2021, Section 3(b)).

Use 4×4 and 16×16 different-skew tables at $n = 1{,}000$ and 10,000, with MI pairs $(0.02, 0.02)$ and $(0.02, 0.03)$. Match the targets and margins to the corresponding

copying populations. In a binary table the two constructions are equivalent; in a larger table copying changes all diagonal and off-diagonal cells, whereas the transfer changes only four. The smaller table checks a representative comparison; the larger table examines whether the sparse-table bias and reversed estimated ordering persist under a different arrangement of association. Compare rejection, validity and estimation diagnostics at every matched setting, including estimated direction under the alternative.

5.3.5 A supporting unequal-sample comparison

A second supporting check changes sample allocation while keeping the total number of observations fixed at 2,000. It uses the copying 4×4 different-skew populations with MI pairs $(0.02, 0.02)$ and $(0.02, 0.03)$ nats. For each pair, compare $(n_P, n_Q) = (1000, 1000), (400, 1600)$ and $(1600, 400)$. Reversing the allocation leaves the populations unchanged: Q retains its more uneven margins and has higher MI in the alternative. This asks whether the same data budget gives different false-positive rates and power when one population has fewer observations.

Unequal sample sizes change both the bias corrections in Equation (3.7) and the weights in the standard error and degrees of freedom. The check applies the same estimators and validity rules to every allocation. Its two balanced comparisons are already part of the main studies, so the unequal allocations add four settings.

A supporting bias-correction check.

A jackknife check changes the MI bias correction while keeping the original standard errors and standard normal distribution cutoff, using the same samples from four existing 16×16 different-skew copying settings. Appendix C.3 specifies the correction and reports all four comparisons.

The complete design contains 70 population pairs and 240 distinct settings. Shared settings, such as the zero-difference points in the power study, use the same simulation results in each comparison. At 20,000 repetitions per setting, the study samples 4,800,000 table pairs.

5.4 Sampling and performance measures

For each setting, draw 20,000 independent multinomial count-table pairs from its fixed P and Q. The samples from the two populations are independent, and both methods are applied to the same pair. The MI estimator, bias correction and

standard error are those in Chapter 3; the Expanded Welch calculation is given in Chapter 4.

All settings use a two-sided 5% test. The primary rejection rate is

$$\frac{\#\{\text{valid p-values no larger than } 0.05\}}{20{,}000}. \tag{5.7}$$

At equal MI this estimates the false-positive rate; when MI differs it estimates power. Invalid calculations contribute no rejection but are not evidence for equal MI, so the valid-result frequency is reported alongside rejection. Rates restricted to method-valid or common-valid samples and their denominators are specified in Appendix B.

Wilson 95% intervals quantify simulation uncertainty for individual rates (Wilson, 1927). Intervals for the difference between methods use paired rejection indicators because the tests receive the same data. For 20,000 repetitions and a rate near 0.05, the Monte Carlo standard error is approximately 0.00154, so departures of only a few thousandths need to be read alongside the intervals. Appendix B.6 gives the calculations.

Diagnostics of the shared estimates.

The known populations also let us measure estimation error. Difference bias is the mean corrected estimate minus the true difference. The sample variance across repetitions measures its observed spread, whereas the mean squared standard error measures the uncertainty assigned by the tests. Dividing the mean squared standard error by the observed variance indicates whether the estimated uncertainty is too large or too small on average. Comparing the observed variance with $V(P)/n_P + V(Q)/n_Q$ separately checks the first-order approximation with its population inputs.

Degrees-of-freedom summaries use the median and 5th–95th percentile range among valid Expanded Welch outputs. This range describes differences between sampled results, not uncertainty in the estimated median. Empty-cell, row and column counts describe sparse sampling. The diagnostics retain finite estimates even if the associated test is invalid; Appendix B specifies the subsets used in each comparison.

Finally, a small timing comparison measures the computational cost on pre-generated 2×2 through 16×16 table pairs at $n = 1{,}000$ under different skew. Sampling and population construction are excluded from the timings. Both methods scan the cells, so their operation counts grow linearly with the number of cells. The timing procedure and measurement environment are given in Appendix B.7.

Chapter 6

Experimental Results and Discussion

The results follow the three experimental questions: when the tests give reliable false-positive rates, how common population MI affects Expanded Welch's estimated degrees of freedom, and their power to detect MI differences. Each simulation rate uses 20,000 sampled table pairs. Blue solid curves show Normal Wald and magenta dashed curves show Expanded Welch. Error bars or bands give pointwise Wilson 95% intervals, and the dotted line marks 0.05. Hollow markers indicate valid-result frequency below 90%. The parameter settings are those in Chapter 5; additional comparisons and the full 120-setting null table are in Appendix C. The chapter ends with a self-contained application to online shopping data.

6.1 False positives across sample size and population structure

6.1.1 Increasing table size

Figure 6.1 holds both population MI values at 0.02 nats and compares the four table sizes under different skew. The dominant marginal probabilities are 0.7 for P and 0.8 for Q, with the remaining mass shared equally across their other categories. Each panel varies only sample size, from 10 to 50,000 per population. The lower strip shows how often each method returns a valid p-value.

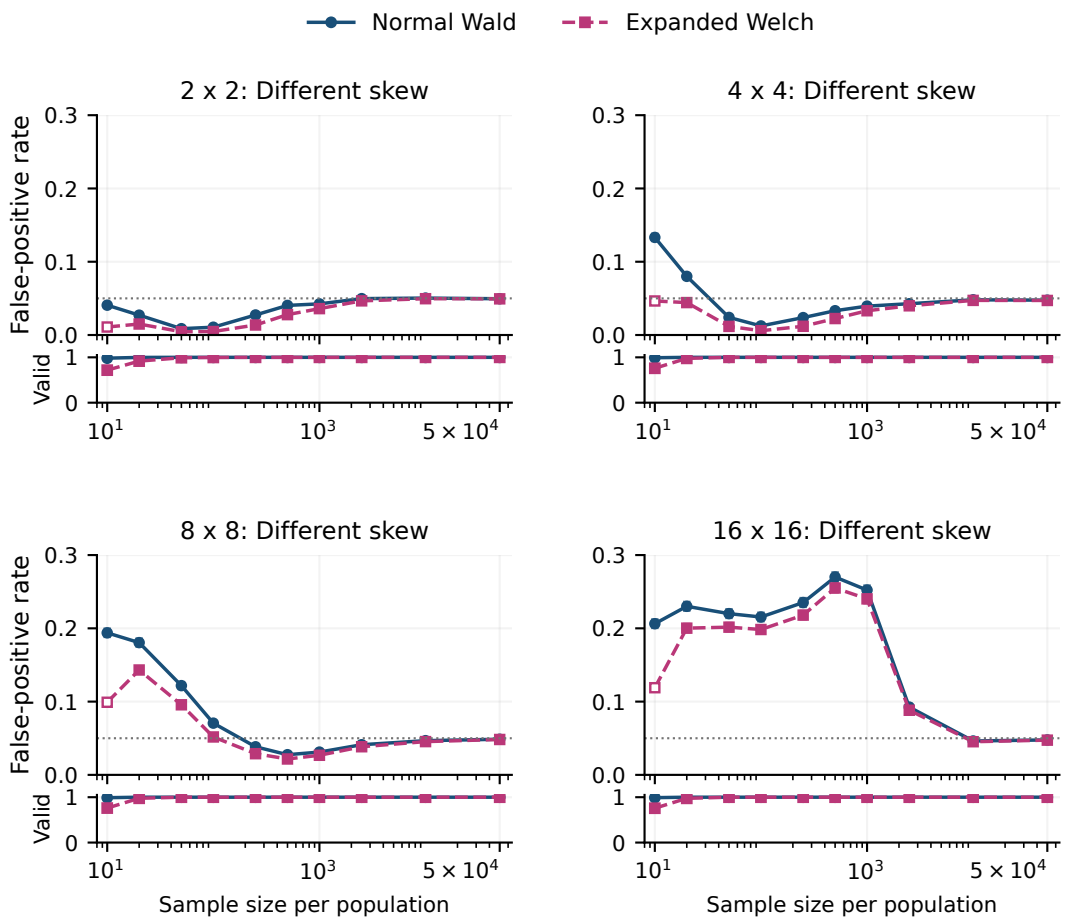


Figure 6.1: False positives and validity under different-skew margins.

The false-positive pattern changes substantially with table size. In 2×2 tables both methods reject too rarely over much of the sample-size range. In 4×4 tables Wald is above 0.05 at $n = 10$ and 20, but both methods fall below the target at larger intermediate sample sizes. In the 8×8 comparison, the transition occurs at larger sample sizes. Expanded Welch lowers rejection throughout, helping where Wald is excessive and moving an already-low rate further below 0.05.

At the smallest sample size, part of that reduction involves lost validity. For 4×4 tables at $n = 10$, Wald rejects at 0.1332 and Expanded Welch at 0.0465, but their valid-result frequencies are 0.9910 and 0.7588. Restricting both tests to the 15,175 common-valid pairs gives rejection rates 0.1029 and 0.0612. The Student t distribution still reduces rejection on those same pairs, but the unconditional rate near 0.05 also reflects failed calculations. Validity and calibration therefore answer different questions.

The 16×16 table shows that valid calculations need not be well calibrated. At $n = 1{,}000$, every calculation is valid, yet Wald rejects at 0.2522 and Expanded Welch at 0.2402. The paired reduction is 0.0120, with approximate 95% interval

[0.0105, 0.0135], so Expanded Welch's larger cutoff lowers rejection but leaves a large excess of false positives. At $n = 10{,}000$ the rates are 0.0463 and 0.0453; both remain near 5% at 50,000. More observations substantially improve this setting. The next comparison asks how much of this pattern changes with the margins.

6.1.2 Changing the margin regime

Figure 6.2 compares all three margin regimes at 4×4 and 16×16, retaining the common MI target and sample-size sweep. Uniform and same-skew populations have identical joint distributions within each pair, whereas different-skew populations have different tables but equal MI. Comparing the columns shows how uneven category frequencies and different population margins change the test.

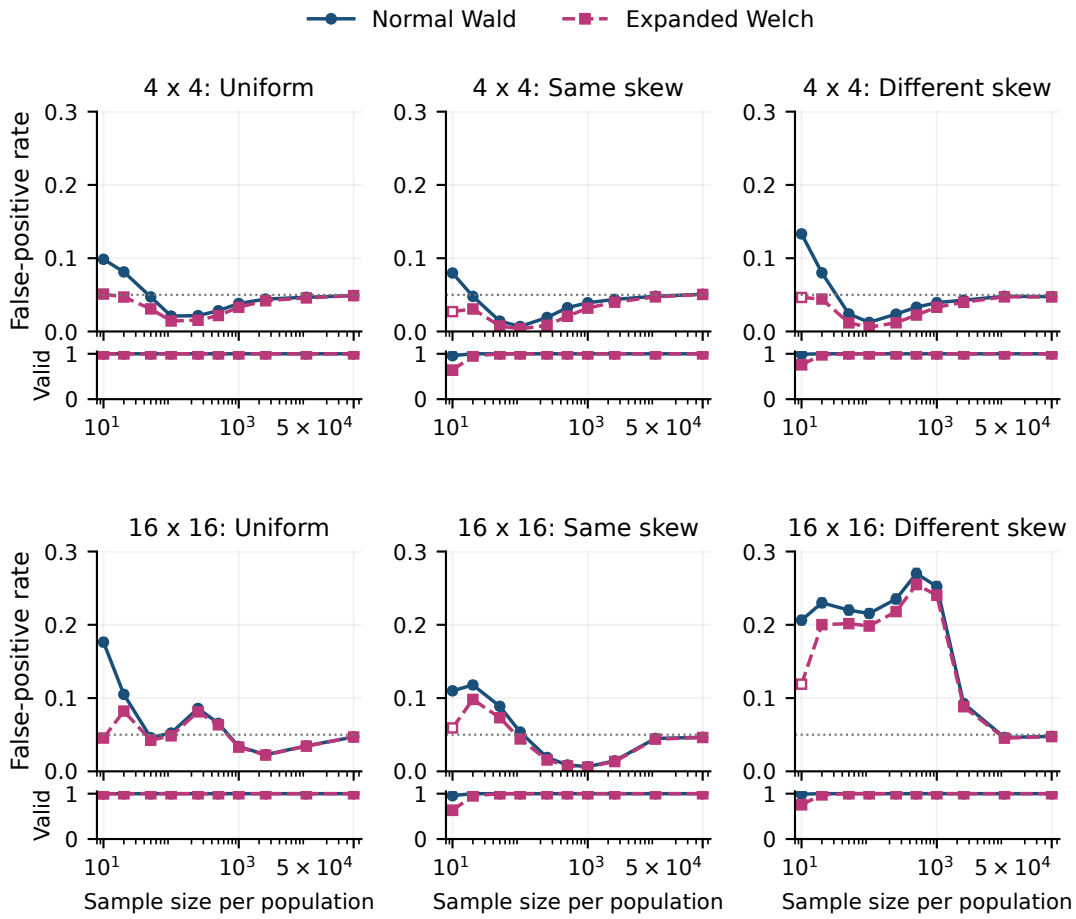


Figure 6.2: False positives and validity across three margin regimes.

At 4×4, all three regimes give low rejection at intermediate sample sizes and rates much nearer 0.05 by $n = 10{,}000$. At $n = 1{,}000$, for example, Wald rates range from 0.0383 to 0.0394, while Expanded Welch ranges from 0.0319 to 0.0331. Changing the margins has a relatively small effect on these particular rates, and

both methods are fully valid. The larger table behaves differently: at $n = 1{,}000$, Wald rejects at 0.0339 under uniform margins, 0.0065 under same skew, and 0.2522 under different skew. Expanded Welch gives 0.0331, 0.0059 and 0.2402, respectively. The different-skew excess is therefore not a general consequence of uneven margins alone.

Excessive false positives also occur when the population tables are identical. With uniform margins, Wald rejects at 0.0737 for 8×8 tables at $n = 100$, and at 0.0859 for 16×16 tables at $n = 250$. All calculations are valid. Because the two samples then have the same distribution and size, their MI estimation biases cancel in the difference. In these settings, the mean squared standard error is approximately 0.84 and 0.77 times the observed variance, respectively. The tests therefore assign too little uncertainty on average, which can increase false positives even without a biased MI difference.

Different population margins introduce a further problem: bias in the estimated MI difference itself. Table 6.1 reports four different-skew null settings. Observed SD is the sample standard deviation of the corrected MI difference. Estimated SD is the square root of the mean squared standard error, rather than the mean standard error. The variance ratio divides the mean squared standard error by the observed sampling variance. All four settings have full validity and finite diagnostics.

Table 6.1: Bias and uncertainty of the estimated MI difference.

Size	n	Bias	Observed SD	Estimated SD	Variance ratio
4×4	1,000	-0.00012	0.01053	0.01085	1.061
4×4	10,000	-0.00001	0.00316	0.00318	1.014
16×16	1,000	0.02960	0.01449	0.02019	1.942
16×16	10,000	-0.00088	0.00387	0.00401	1.073

The 16×16, $n = 1{,}000$ difference has bias 0.02960 nats, more than twice its observed SD of approximately 0.01449. To see where this shift comes from, recall that bias is the mean estimate across samples minus the population value. The bias of a difference is exactly bias of P's MI estimate minus the bias of Q's MI estimate. Here the individual corrected MI biases are approximately -0.00794 for P and -0.03754 for Q. Since the true difference is zero, the estimated difference is centred at

$$\mathbb{E}(\widehat{\Delta}_{\mathrm{BC}}) \simeq 0 + (-0.00794) - (-0.03754) = 0.02960 \quad \text{nats.} \tag{6.1}$$

Both corrected estimates are too low on average, but Q's is lower by more, so subtracting it creates a positive difference even though the population MI values

match. The fixed-alphabet bias correction cancels because the table and sample sizes match, so the raw plug-in difference has exactly the same bias. The sparse-sample bias in the two MI estimates is not captured by their common leading correction (Paninski, 2003). In this copying setting, an average of 147 of P's 256 cells and 183 of Q's cells are empty.

For the copying 16×16, $n = 1{,}000$ pair, estimated uncertainty is also inaccurate: mean squared standard error is about 1.94 times the observed variance. Excessive false positives nevertheless remain: overestimating the variance does not remove the positive bias in the estimated MI difference. At $n = 10{,}000$ the bias is much smaller and the variance ratio is 1.07, accompanying the improved null rate. The population first-order variance is about 0.60 times the observed variance at $n = 1{,}000$, compared with 0.84 at 10,000, showing that the approximation itself also needs attention.

The four-cell construction gives a similar failure at the matched 16×16, $n = 1{,}000$ null setting. Its difference bias is 0.03145 nats, and the false-positive rates are 0.3095 for Wald and 0.2977 for Expanded Welch, with every calculation valid. The shifted difference and excessive rejection therefore occur under both tested arrangements of association. At $n = 10{,}000$, the four-cell rates are 0.0499 and 0.0491, and its difference bias is much smaller. Table C.8 gives the full matched comparison. Expanded Welch leaves the shared estimates unchanged; the next study examines how common population MI affects its estimated degrees of freedom and rejection cutoff.

6.2 Population MI and degrees of freedom

Figure 6.3 keeps 4×4 different-skew tables and varies the common population MI from 0.0001 to 0.05 nats. The two columns use $n = 100$ and 1,000. Upper panels show false positives; lower panels show the median estimated combined degrees of freedom, with the 5th–95th percentile range shaded. Every calculation in this sweep is valid, so the low rejection rates do not reflect missing test outputs.

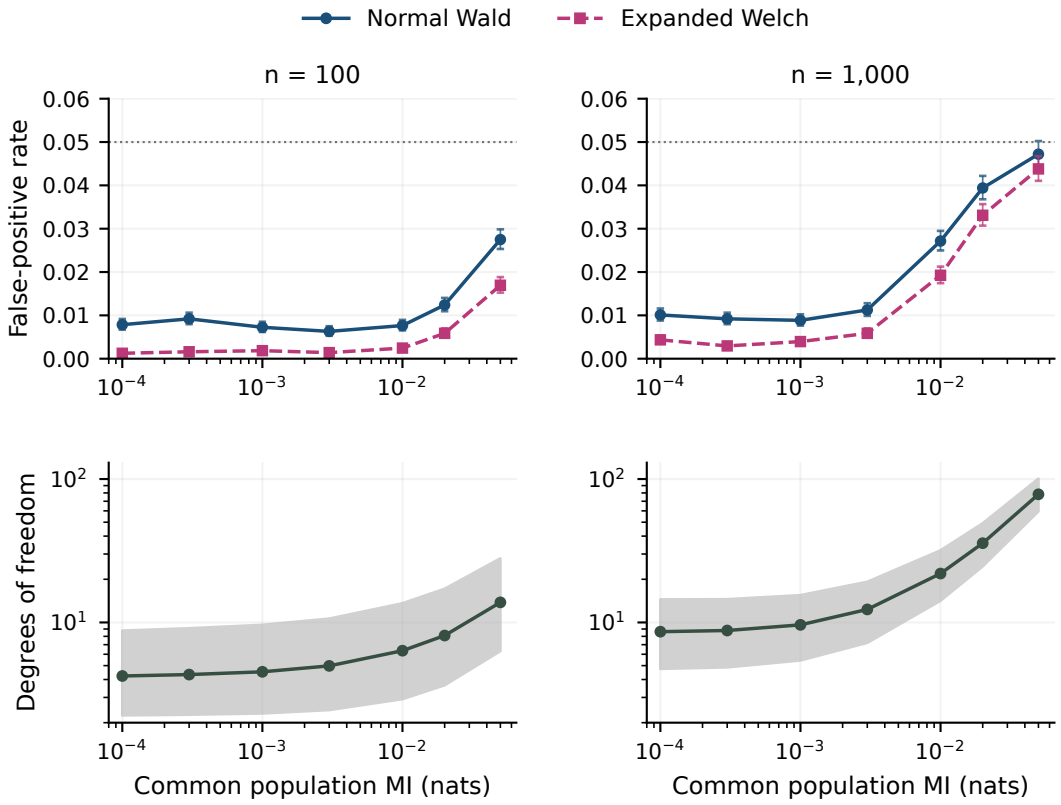


Figure 6.3: Common population MI, false positives and estimated degrees of freedom.

Both tests reject too rarely near independence, and Expanded Welch rejects still less. At common MI 0.0001 and $n = 1{,}000$, Wald's rate is 0.0101 and Expanded Welch's is 0.0044. At common MI 0.05, Wald rejects at 0.0472 and Expanded Welch at 0.0438. Larger common MI brings the rates closer to 0.05 in this comparison, especially at the larger sample size. The $n = 100$ rates remain low even at the largest MI target, so a larger MI value does not substitute for enough observations.

Estimated degrees of freedom increase with common population MI. At $n = 1{,}000$, their median rises from 8.62 at MI 0.0001 to 78.24 at MI 0.05; at $n = 100$, it rises from 4.23 to 13.80. Smaller degrees of freedom give a larger cutoff from the Student t distribution, requiring a larger absolute statistic for rejection. The population approximation in Section 4.5 describes a table approaching independence, whereas the estimated degrees of freedom use a sampled table. Sampling noise creates apparent MI even when population MI is almost zero, so the estimated degrees of freedom remain several units rather than following the population approximation down to zero. Appendix A.4 gives the corresponding quadratic calculation. Expanded Welch still requires a larger statistic for rejection than Wald, although Wald already rejects too rarely.

The median degrees of freedom illustrate the change in cutoff. For a two-sided 5% test, the cutoffs from the Student t distribution are approximately 2.277 at 8.62 degrees of freedom and 1.991 at 78.24, compared with Normal Wald's 1.960. Thus $|T| = 2.1$ would reject under Wald but not under Expanded Welch with 8.62 degrees of freedom; at 78.24 both would reject. These examples use the reported medians, whereas each sampled table pair uses its own estimated degrees of freedom.

The predicted sampling variability of the estimated MI variance can also be checked directly. Table C.4 compares the measured variation of $\widehat{V}(P)$ across samples with the population prediction $\tau^2(P)/n_P$ and the average plug-in prediction $\widehat{\tau}^2(P)/n_P$, and repeats the comparison for Q. In 4×4 tables at common MI 0.02 and $n = 10{,}000$, both predictions are within 2% of the observed variability. At common MI 0.0001 and $n = 1{,}000$, the population prediction accounts for only about 5% of that variability, while the average plug-in prediction is 1.95 times the observed value for P and 2.09 times for Q. Near independence, predictions of the MI variance estimate's sampling variability therefore have substantial errors with both population and empirical inputs.

Errors also arise in the shared MI statistic. At MI 0.0001 and $n = 1{,}000$, mean squared standard error is 1.74 times the observed variance. The population first-order variance is only about 0.04 times that variance, indicating that the first-order approximation is poor this close to independence at this sample size. These ratios compare the population first-order variance and the plug-in estimate with observed sampling variance. The population first-order variance uses the true probabilities in $V(P)/n_P + V(Q)/n_Q$, whereas the plug-in variance estimate replaces them by each sample's observed proportions before averaging the resulting squared standard errors. They need not have the same error.

Near independence, the true first-order variance is small because the population cell probabilities are close to the products of their margins. A finite sample can still show apparent association: observed cell proportions generally differ from the products of their observed margins. The plug-in variance estimate treats those departures as features of an estimated population table and can therefore assign a substantially larger first-order variance. Meanwhile, the linear approximation at the true table omits the quadratic sampling terms that matter near independence. Brillinger (2004) distinguishes these first-order and independence calculations; Appendix A.4 shows why the quadratic terms remain when the first-order term vanishes. Thus a population first-order variance that is too small can coexist with a plug-in variance estimate that is too large. The observed ratios show both errors in this setting.

At MI 0.05, mean squared standard error is approximately 1.00 times the observed variance, and the population first-order variance is 0.95 times the observed variance. Detailed values are in Appendix C. Changing the rejection cutoff cannot correct errors in the estimated MI difference or standard error. The next study examines power when the populations have different MI values.

6.3 Power to detect MI differences

6.3.1 Power across table sizes

Figure 6.4 fixes $I(P) = 0.02$ nats, sets $I(Q) = 0.02 + |\Delta|$, and uses different-skew margins. Each panel compares $n = 1{,}000$ and 10,000 at one table size. The figure enlarges differences up to 0.02 nats; the full grid through 0.2 is in Figure C.2. Every displayed setting is fully valid.

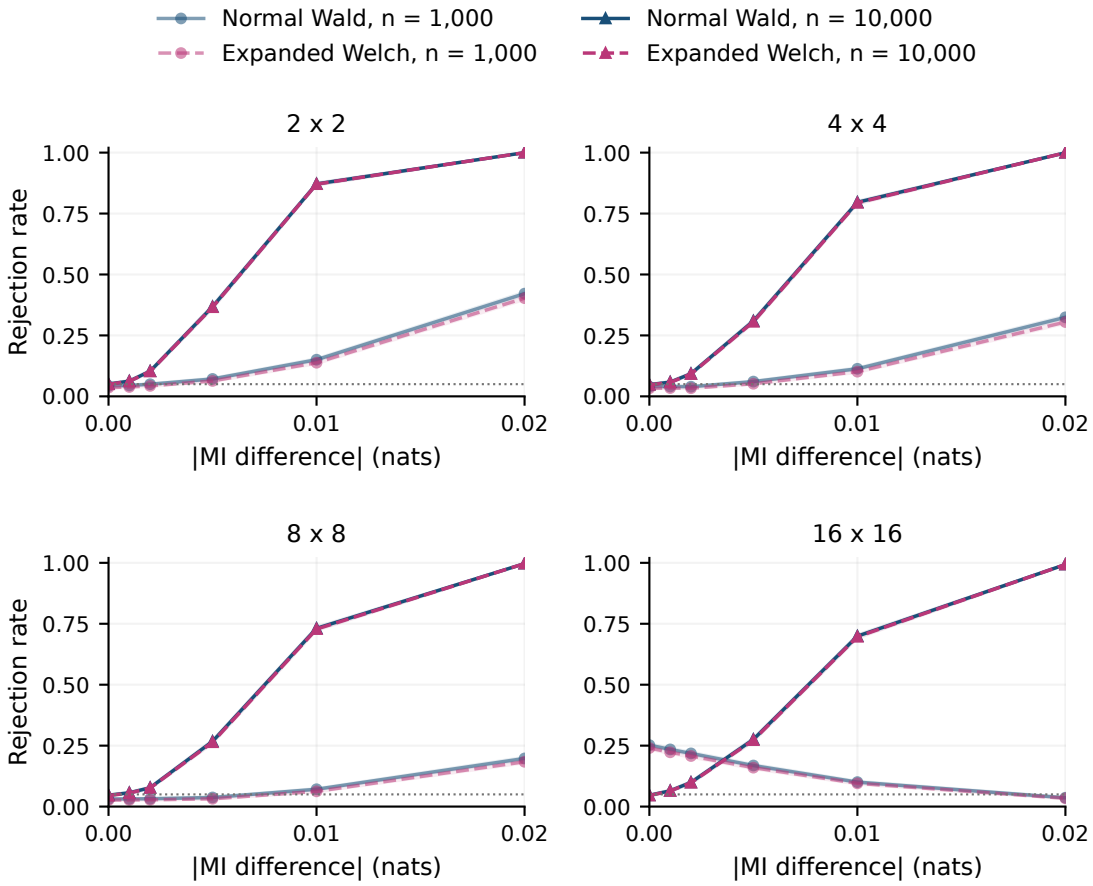


Figure 6.4: Rejection rates over small MI differences under different-skew margins.

For the smaller tables, increasing the magnitude of the true MI difference generally increases power, and larger samples make a clear improvement. In 4×4

tables, Wald and Expanded Welch have estimated power of 0.1132 and 0.1013 for a 0.01-nat difference at $n = 1{,}000$, compared with 0.7962 and 0.7940 at $n = 10{,}000$. Expanded Welch's larger cutoff costs some power at the smaller sample size and very little at the larger one, because higher estimated degrees of freedom bring Expanded Welch's cutoff closer to Wald's. The null points remain essential: these curves compare tests at the same nominal threshold, not at equal achieved false-positive rates.

The 16×16, $n = 1{,}000$ curve instead falls initially as the magnitude of the true difference grows. Wald rejects at 0.2522 under equal MI, 0.1005 at a difference of 0.01, and 0.0363 at 0.02. The positive bias identified earlier opposes the negative true difference. The mean estimated difference is approximately 0.02960 at the null. Under the next two alternatives, the true difference and estimated difference bias combine as follows:

$$\mathbb{E}(\widehat{\Delta}_{\mathrm{BC}}) \simeq -0.01 + 0.03061 = 0.02061, \tag{6.2}$$

$$\mathbb{E}(\widehat{\Delta}_{\mathrm{BC}}) \simeq -0.02 + 0.03164 = 0.01164. \tag{6.3}$$

The bias in the estimated MI difference changes slightly across these alternatives, but remains positive and larger than the magnitude of the true difference. As the true difference becomes more negative, it offsets more of that bias, so the centre of the estimated difference moves towards zero rather than away from it. Dividing by the estimated standard error can then give smaller absolute statistics, despite the larger population difference. Unequal estimation bias therefore makes rejection fall as the magnitude of the true difference grows over this range. The full-range comparison shows rejection rising again at larger differences.

The bias can also reverse the estimated direction of the difference. Although Q has the larger population MI at $|\Delta| = 0.01$, 2,009 of the 2,010 Wald rejections and 1,905 of the 1,906 Expanded Welch rejections have a positive estimated difference, indicating larger MI for P. The rejection curves count all two-sided rejections; they do not show whether the estimated ordering is correct. Table C.6 separates these outcomes.

At $n = 10{,}000$, the 16×16 bias is much smaller, and power increases across the small differences. For a 0.01-nat difference, Wald's estimated power is 0.6992 and Expanded Welch's is 0.6968; by 0.02 the rates are 0.9932 and 0.9931. Every rejection in these two larger-sample comparisons has the correct negative estimated direction. The larger sample improves both the null behaviour and power.

Under the matched four-cell construction at $n = 1{,}000$, 2,493 of 2,494 Wald rejections and 2,350 of 2,351 Expanded Welch rejections also indicate the opposite

ordering. Concentrating association in four cells therefore does not remove the direction error at this sample size. At $n = 10{,}000$, every rejection under either construction has the correct estimated direction, as shown in Table C.8.

The jackknife check asks whether changing the MI bias correction can improve the estimated ordering without changing the rejection cutoff. Replacing Wald's analytic MI bias correction with a jackknife correction reduces wrong-direction rejections from 2,009 of 2,010 to 5 of 549 for the 0.01-nat alternative at $n = 1{,}000$. Correct-direction rejections increase from 1 to 544. The check retains the existing standard error and standard normal distribution cutoff, and its null rejection rate of 0.0085 remains below the 0.05 target. Appendix C.3 reports this four-setting check.

6.3.2 Power across margin regimes

Figure 6.5 fixes the table size at 4×4, sets $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$, and compares the three margin regimes at $n = 1{,}000$ and 10,000. This isolates the effect of the margin regime on power at a fixed table size.

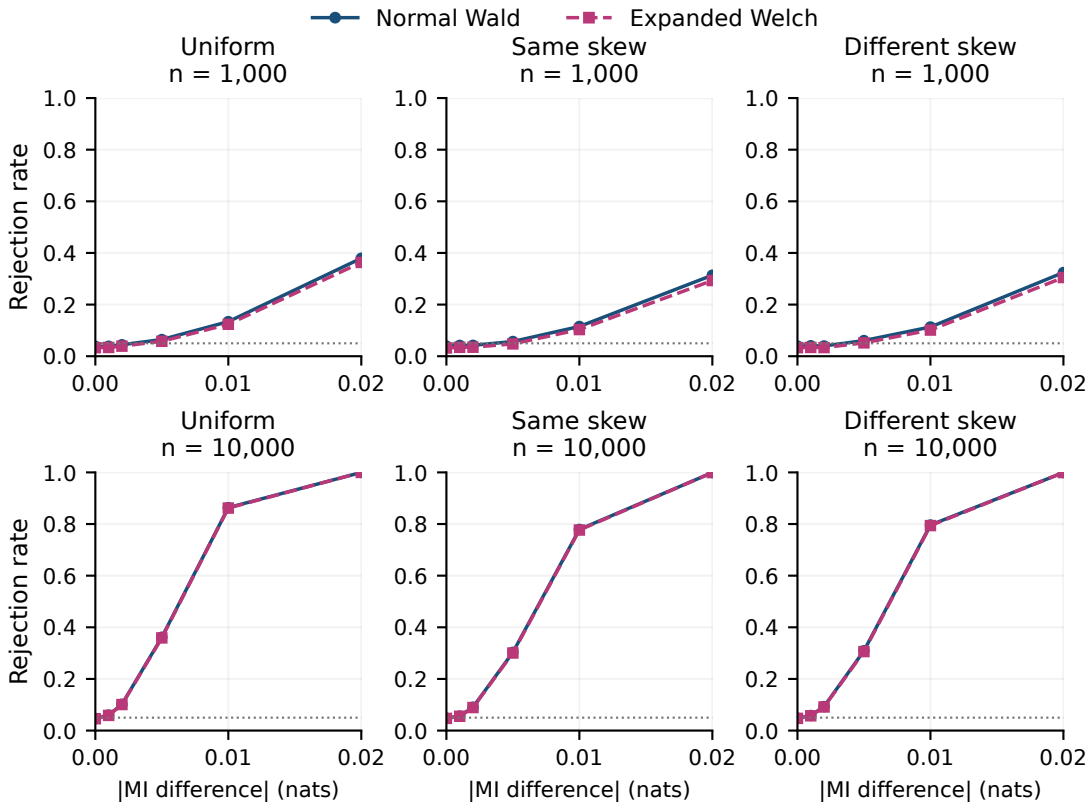


Figure 6.5: Rejection rates over small MI differences in 4×4 tables.

Uniform margins give somewhat higher power in these settings. At $n = 10{,}000$ and a 0.01-nat difference, Wald's estimated power is 0.8635 under uniform margins,

0.7790 under same skew, and 0.7962 under different skew. Expanded Welch gives 0.8619, 0.7768 and 0.7940. The margin-related change is larger than the difference between tests at this sample size.

The observed precision helps explain the ordering of power. For this 0.01-nat difference at $n = 10{,}000$, the sample SD of the estimated difference is approximately 0.00327 nats under uniform margins, 0.00364 under same skew and 0.00361 under different skew. The biases are small in all three settings, and the square roots of the mean squared standard errors are close to these observed SDs. The same population difference is therefore easier to distinguish from sampling variation under the uniform construction. Equal MI targets do not imply equal precision: the cell probabilities also determine the variance in Equation (2.30).

At $n = 1{,}000$, Expanded Welch's larger cutoff produces a more visible reduction in power, while all three null points are already below 0.05. By a difference of 0.2 nats, both methods reject every simulated pair in all three regimes at both sample sizes.

The supporting four-cell check gives the same qualitative comparison at its matched 4×4 settings: Expanded Welch lowers an already-low null rate at $n = 1{,}000$, and changes power little at 10,000. Table C.7 gives the rates and diagnostics for both constructions.

The unequal-sample check keeps these 4×4 different-skew populations fixed and changes the allocation of 2,000 observations. For the 0.01-nat difference, giving Q only 400 observations reduces power to 0.0540 for Wald and 0.0340 for Expanded Welch, compared with their balanced rates of 0.1132 and 0.1013. All calculations are valid. The lower degrees of freedom under imbalance increase Expanded Welch's rejection cutoff, moving already-conservative false-positive rates further below 5%. Table C.9 reports both allocation directions.

6.4 Computational cost

Table 6.2 compares time per test call on different-skew null table pairs at $n = 1{,}000$. The timings exclude sampling and population construction. Appendix B.7 gives the measurement procedure and computing environment.

Expanded Welch costs approximately 1.8–1.9 times as much in these timings. Both methods require $O(k^2)$ work for a $k \times k$ table, but fixed call overhead can be important for these small arrays. Median times remain below 115 microseconds for Wald and 205 microseconds for Expanded Welch across the four table sizes.

Table 6.2: Time per test call in microseconds: median [interquartile range].

Size	Normal Wald	Expanded Welch
2×2	101.83 [99.25, 104.87]	186.58 [182.46, 191.67]
4×4	107.85 [101.92, 113.51]	197.13 [188.83, 205.25]
8×8	103.46 [101.12, 107.30]	191.81 [187.00, 197.26]
16×16	112.42 [108.33, 116.43]	202.29 [198.07, 208.00]

6.5 Application to online shopping data

The Online Shoppers Purchasing Intention dataset provides a practical comparison of month–purchase MI between new and returning visitors (Sakar and Kastro, 2018). A larger MI means that the month provides more information about whether a session ends in a purchase. A second comparison uses the weekend indicator instead of month, giving a contrasting association in the same dataset.

The dataset contains 12,330 sessions, with each session belonging to a different user according to the source documentation (Sakar and Kastro, 2018). The analysis retains all 1,694 sessions labelled `New_Visitor` and 10,551 labelled `Returning_Visitor`, excluding the 85 `Other` sessions. The outcome is the original Boolean `Revenue` variable: true indicates a purchase and false indicates no purchase. The tests treat the two disjoint visitor groups as independent samples and the sessions within each group as independent observations.

Each group forms a 10×2 month–purchase count table. The rows are the ten months recorded in the dataset: February, March, May, June, July, August, September, October, November and December. The columns are no purchase and purchase, and each session adds one to its corresponding cell. Both groups use these same categories, with no smoothing or removal of zero cells. All ten months occur in both groups, although the new-visitor February row contains only one session and no purchase. Replacing month with the Boolean `Weekend` variable gives a 2×2 table for each group, with weekday and weekend rows; all four cells are observed.

Both methods use the bias-corrected MI estimates in Table 6.3. For the month tables, the leading bias correction subtracts $(10-1)(2-1)/(2n) = 9/(2n)$ from each plug-in MI estimate (Miller, 1955; Moddemeijer, 1989); for the weekend tables, it subtracts $1/(2n)$. Here n is the number of sessions in the relevant visitor group. The estimated difference is new-visitor MI minus returning-visitor MI for the same predictor–purchase pair. Normal Wald compares this difference divided by its standard error with the standard normal distribution; Expanded Welch uses the

same statistic with a Student t distribution and estimated degrees of freedom. Both tests are two-sided at the 5% level.

Table 6.3: MI estimates and two-sample test results for online shopping sessions.

	Corrected MI (nats)			p-value	
Predictor	New	Returning	Expanded Welch degrees of freedom	Normal Wald	Expanded Welch
Month	0.004316	0.017880	26.79	4.71×10^{-5}	3.73×10^{-4}
Weekend	0.000664	0.000727	2.18	0.9554	0.9601

Month has higher estimated MI with purchasing among returning visitors. The corrected difference is -0.013563 nats, with shared standard error 0.003333 nats, giving

$$T \approx \frac{-0.013563}{0.003333} \approx -4.069. \tag{6.4}$$

With 26.79 estimated degrees of freedom, the Student t distribution gives a larger p-value than the standard normal distribution, but both tests reject equal month–purchase MI for new and returning visitors. This application was selected from an initial exploration of six comparisons across three public datasets. Accounting for all six comparisons with Holm's multiple-testing adjustment (Holm, 1979) gives month p-values of 0.000189 for Normal Wald and 0.00149 for Expanded Welch, so the conclusion remains unchanged. Table C.10 lists all six comparisons and their raw and adjusted p-values.

The weekend comparison gives a different result: the corrected weekend–purchase MI estimates for new and returning visitors are both approximately 0.0007 nats. Their difference, -0.000063 nats, is small relative to the shared standard error of 0.001124 nats, giving $T \approx -0.056$. Although Expanded Welch estimates only 2.18 degrees of freedom, both p-values remain close to one. Neither test provides evidence that weekend–purchase MI differs between the visitor groups, but this does not establish that their population MI values are equal. The small statistic, rather than the degrees of freedom alone, explains the agreement between the methods here.

Expanded Welch gives larger p-values than Normal Wald while preserving the month–purchase and weekend–purchase MI differences and, here, both test decisions. Its degrees of freedom depend on the relative variability of the MI variance estimates, so large session counts alone need not make its p-values close to Normal Wald's. These comparisons describe the month–purchase and weekend–purchase associations in the recorded sessions.

Chapter 7

Implications and Limitations

7.1 Implications for using Expanded Welch

Expanded Welch improves false-positive control in several small-sample settings. For uniform 4×4 populations at $n = 20$, it lowers the false-positive rate from 0.0813 to 0.0470, with every calculation valid. For 8×8 different-skew populations at $n = 100$, it lowers the rate from 0.0706 to 0.0518, again with full validity.

The power comparison uses larger samples, at $n = 1{,}000$ and 10,000, so it does not establish the power cost in those smaller-sample settings. Across the power comparisons, Expanded Welch rejects less often and offers no general advantage over Normal Wald. It also lowers false-positive rates that are already conservative and does not correct the unequal biases in the MI difference. Normal Wald therefore remains the simpler baseline. Valid-result frequency remains important, because a lower rejection rate can otherwise reflect failed calculations.

For an observed feature–label comparison, report the estimated MI difference, its standard error and the p-value together. The difference gives the estimated size and direction of the change, whereas the standard error gives the sampling uncertainty assigned by the method. The sparse-table results show why the direction matters: unequal estimation biases can shift or reverse the estimated difference, as Equations (6.1) and (6.3) illustrate. If Expanded Welch is used, report its estimated degrees of freedom as well, since they determine the rejection cutoff for that comparison.

Before applying either test, assess its performance on populations that match the application's category definitions, sample sizes and margin patterns. Include plausible population MI values and distinct equal-MI tables, then compare false positives, power and usable-result frequency. Because neither method is reliable in every studied setting, this assessment is particularly important for sparse data.

Choose the method using that evidence before examining the application's observed p-values.

7.2 Limits of the evidence

The rejection rates and Monte Carlo intervals apply to the constructed population pairs. The intervals quantify uncertainty from the finite number of simulated samples; they do not describe how performance varies across untested populations.

The primary populations use a copying mixture with identical row and column margins within each population, allowing association to vary without changing category frequencies. The four-cell check supports the representative 4×4 method comparison and reproduces the main 16×16 estimation failures at its matched settings, although power changes numerically with the construction. Copying favours matching categories in both populations, whereas the four-cell transfer concentrates association in the same chosen block. The different-skew regime changes category frequencies while keeping those association locations aligned. Its results therefore demonstrate that distinct equal-MI tables can be difficult to compare, but do not describe every way two tables with equal MI can differ. A broader assessment would vary association locations, population MI values and margins, including unequal row and column margins within a population.

Because equal sample sizes make the two leading bias corrections cancel in $\widehat{\Delta}_{\mathrm{BC}}$, the remaining difference bias in the main studies is exactly the bias of the raw plug-in MI difference. This conclusion concerns the fixed-alphabet correction used here. With unequal sample sizes, the corrections no longer cancel, and the two contributions to the standard error and degrees of freedom can become less balanced. The derivation allows unequal sample sizes. The supporting fixed-total check assesses both allocation directions in two 4×4 different-skew pairs. There, imbalance changes bias, uncertainty and power, while the larger Expanded Welch cutoff lowers already-conservative null rates. This extends the comparison beyond equal samples, but a wider allocation study would be needed for other table sizes and population structures.

The sampling assumptions also limit transfer to other applications. Paired observations would require a covariance term in the variance of the difference, whereas the present samples are independent. Continuous MI estimators have different estimation errors from the categorical MI estimator used here. Similarly, increasing the number of categories with sample size changes the fixed-alphabet setting of the derivation. The empirical recommendation concerns the independent

categorical comparisons studied here, at a two-sided 5% threshold. The online-shopping application illustrates interpretation on observed data, but its unknown population MI values do not allow a direct measurement of false-positive rates or power.

Exact independence needs a different mathematical approximation because the first-order MI variance vanishes there. The population-MI study also shows that positive population MI is not enough to make that approximation accurate at a particular sample size: small positive MI can remain close to the problematic region. This explains why improving the rejection cutoff and improving MI's sampling approximation near independence are separate tasks. Appendix A.4 derives the second-order approximation needed there.

Chapter 8

Conclusion

8.1 Contributions and main findings

This thesis derives an Expanded Welch–Satterthwaite test for comparing MI of the same pair of categorical variables across independent populations. Its mathematical contribution is the observation-level sensitivity of the estimated MI variance, which determines the degrees of freedom. Its empirical contribution is a controlled 240-setting comparison with Normal Wald that identifies the estimation errors behind the observed false-positive rates and power. Three connected studies are supported by direct checks of variance-estimate variability, population construction and sample allocation. An online-shopping application illustrates the interpretation of both tests for observed categorical data.

Direct checks of the variance-sensitivity formula show close agreement between predicted and observed variability of the MI variance estimate in the well-sampled 4×4 setting, but substantial approximation and plug-in errors near independence. This identifies where the first-order approximation describes the sampling variability of the MI variance estimate and where it needs further development.

Expanded Welch improves false-positive control in some small-sample settings, including uniform 4×4 populations at $n = 20$, but the power cost in those settings remains unmeasured. Near independence it makes conservative rates lower still. Other sparse settings expose errors in the shared estimates: uncertainty can be underestimated even for identical populations, while unequal MI biases in different-skew tables can shift or reverse the estimated difference. The bias and ordering failures occur under both tested population constructions. Larger samples improve the estimation diagnostics and observed performance, and the tests become similar. These findings do not support Expanded Welch as a general replacement for Normal Wald. Normal Wald remains the simpler baseline, although sparse estimation can

make neither test reliable. The main practical requirement is accurate estimation of the MI difference and its uncertainty, which a larger cutoff cannot supply.

8.2 Limitations

The recommendation is limited to independent categorical samples with fixed categories. The controlled populations and small allocation check do not cover every association pattern or sample-size imbalance. The first-order derivation requires positive cell probabilities and positive first-order MI variance; exact independence needs a different approximation.

8.3 Future work

Future work should first address bias in sparse MI differences and the dependence between the difference and its estimated standard error. The 16×16 different-skew jackknife check provides a starting point: at $n = 1{,}000$, it improves the estimated direction while the test remains conservative. Further comparisons should assess changes to the MI difference and its standard error separately before combining them. The falling rejection curve gives a concrete target: in these sparse settings, an improved comparison should avoid moving the estimated difference towards zero as the magnitude of the population MI difference increases. Alternative bias treatments should therefore be assessed through both null calibration and the shape of rejection curves, rather than through MI estimation error alone.

Near independence, a second-order approximation to MI's sampling variation may be more appropriate than changing the rejection cutoff of the first-order test. A likelihood-ratio or resampling method would need validation for equal MI even when the complete population tables differ. This would address the approximation problem revealed by the population-MI diagnostics. Broader unequal-sample comparisons, other association patterns and applications to additional datasets would extend the practical reach of these conclusions.

Appendix A

Supporting Calculations

These calculations support the variance sensitivity, near-independence interpretation and validity conditions in Chapter 4.

A.1 Mean of the variance influence function

The observation-level sensitivities $g_P(x, y)$ have a probability-weighted mean of zero. Averaging the one-cell perturbation directions leaves the population table unchanged:

$$\sum_{x,y} p_{xy}\{\delta_{(x,y)} - P\} = P - P = 0. \tag{A.1}$$

This weighted cancellation also holds after exact plug-in substitution. Because the first derivative is linear in the direction of change, the weighted average of the one-cell derivatives is the derivative in the weighted-average direction:

$$\begin{aligned} \mathrm{E}_P\{g_P(X, Y)\} &= \sum_{x,y} p_{xy} \frac{\mathrm{d}}{\mathrm{d}\varepsilon} V\{(1-\varepsilon)P + \varepsilon\delta_{(x,y)}\}\Big|_0 && \text{(A.2)} \\ &= \frac{\mathrm{d}}{\mathrm{d}\varepsilon} V\left\{P + \varepsilon \sum_{x,y} p_{xy}\{\delta_{(x,y)} - P\}\right\}\Big|_0 && \text{(A.3)} \\ &= \frac{\mathrm{d}}{\mathrm{d}\varepsilon} V(P)\Big|_0 = 0. && \text{(A.4)} \end{aligned}$$

A.2 Fixed-margin behaviour near independence

Let $P_0 = (q_{ij})$ be independent, with positive cells $q_{ij} = a_i b_j$. Consider $P_t = P_0 + tH$, where every row and column sum of H is zero. For sufficiently small t, all cells remain positive and the margins remain a and b. We expand around $t = 0$, where $I(P_0) = 0$.

For $f(t) = I(P_t)$, the second-order Taylor formula is

$$f(t) = f(0) + f'(0)t + \frac{f''(0)}{2!}t^2 + O(t^3). \tag{A.5}$$

Here $O(t^3)$ collects the omitted terms whose magnitude is bounded by a constant times $|t|^3$ as t approaches zero. These terms become smaller than the retained quadratic term. The same notation below describes the size of each omitted remainder near zero.

The fixed margins make the pointwise MI in cell (i, j)

$$\ell_{P_t}(i,j) = \log\left(\frac{q_{ij} + tH_{ij}}{q_{ij}}\right) = \log\left(1 + \frac{tH_{ij}}{q_{ij}}\right) = \frac{tH_{ij}}{q_{ij}} - \frac{t^2H_{ij}^2}{2q_{ij}^2} + O(t^3), \tag{A.6}$$

using $\log(1 + z) = z - z^2/2 + O(z^3)$ with $z = tH_{ij}/q_{ij}$. Multiplying by the cell probability gives

$$\begin{aligned}(q_{ij} + tH_{ij})\ell_{P_t}(i,j) &= (q_{ij} + tH_{ij})\left\{\frac{tH_{ij}}{q_{ij}} - \frac{t^2H_{ij}^2}{2q_{ij}^2} + O(t^3)\right\} \\ &= tH_{ij} - \frac{t^2H_{ij}^2}{2q_{ij}} + \frac{t^2H_{ij}^2}{q_{ij}} + O(t^3) \\ &= tH_{ij} + \frac{t^2H_{ij}^2}{2q_{ij}} + O(t^3).\end{aligned}$$

Multiplication by q_{ij} supplies the negative quadratic term; multiplication by tH_{ij} supplies the positive one. Their combination leaves half the positive term. When we sum over cells, the first-order terms cancel because $\sum_{i,j} H_{ij} = 0$. Thus $f(0) = f'(0) = 0$, $f''(0) = \sum_{i,j} H_{ij}^2/q_{ij}$, and

$$I(P_t) = \frac{t^2}{2}\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3). \tag{A.7}$$

To calculate $V(P_t)$, first square the pointwise-MI expansion:

$$\ell_{P_t}(i,j)^2 = \left\{\frac{tH_{ij}}{q_{ij}} + O(t^2)\right\}^2 = \frac{t^2H_{ij}^2}{q_{ij}^2} + O(t^3).$$

Multiplying by the cell probabilities and summing gives its second moment:

$$\begin{aligned}\mathrm{E}_{P_t}\{\ell_{P_t}(X,Y)^2\} &= \sum_{i,j}(q_{ij} + tH_{ij})\left\{\frac{t^2H_{ij}^2}{q_{ij}^2} + O(t^3)\right\} \\ &= t^2\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3).\end{aligned}$$

MI itself begins with a quadratic term, so $I(P_t)^2 = O(t^4)$. Subtracting it from this second moment leaves the quadratic term unchanged:

$$V(P_t) = t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \tag{A.8}$$

The row conditional mean also has no first-order term. Its denominator remains a_i because the margins are fixed, and

$$\begin{aligned} \mathrm{E}_{P_t}\{\ell_{P_t}(X,Y) \mid X = i\} &= \frac{1}{a_i} \sum_j (q_{ij} + tH_{ij}) \left\{ \frac{tH_{ij}}{q_{ij}} + O(t^2) \right\} \\ &= \frac{t}{a_i} \sum_j H_{ij} + O(t^2) = O(t^2). \end{aligned}$$

The last equality uses the zero row sum of H. The column mean has the same cancellation, using $\sum_i H_{ij} = 0$.

In Equation (4.29), the squared centred pointwise MI, $V(P_t)$, the conditional means and $I(P_t)$ all begin at order t^2. Only the unsquared pointwise-MI term in the bracket has a first-order contribution. Therefore Equation (4.29) supplies the first-order term:

$$g_{P_t}(i,j) = \frac{2tH_{ij}}{q_{ij}} + O(t^2). \tag{A.9}$$

Since g_{P_t} has mean zero, its variance is its second moment:

$$\begin{aligned} \tau^2(P_t) &= \sum_{i,j} (q_{ij} + tH_{ij}) \left\{ \frac{2tH_{ij}}{q_{ij}} + O(t^2) \right\}^2 \\ &= 4t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \end{aligned} \tag{A.10}$$

Comparing the leading terms of $I(P_t)$, $V(P_t)$ and $\tau^2(P_t)$ gives $V(P_t) \approx 2I(P_t)$ and $\tau^2(P_t) \approx 4V(P_t)$ for small positive MI along this path. At $t = 0$, the first-order sampling variance is zero; Appendix A.4 treats that case separately.

A.3 Dependence between the MI and variance estimates

Both $\widehat{I}(P)$ and $\widehat{V}(P)$ are calculated from the same sample. Under fixed positive support, their first-order covariance is

$$\mathrm{Cov}\{\widehat{I}(P), \widehat{V}(P)\} \approx \frac{1}{n_P} \mathrm{E}_P[\{\ell_P(X,Y) - I(P)\} g_P(X,Y)]\,. \tag{A.11}$$

The corresponding expression for Q replaces P and n_P with Q and n_Q. To determine how the MI and variance estimates move together near independence, use the fixed-margin path $P_t = (q_{ij} + tH_{ij})$ from Appendix A.2. For small positive t, the expansions there give

$$\ell_{P_t}(i,j) - I(P_t) = \frac{tH_{ij}}{q_{ij}} + O(t^2), \qquad g_{P_t}(i,j) = \frac{2tH_{ij}}{q_{ij}} + O(t^2). \tag{A.12}$$

Substituting both expressions into the first-order covariance above gives

$$\mathrm{Cov}\{\widehat{I}(P_t), \widehat{V}(P_t)\} \approx \frac{2t^2}{n_P} \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O\left(\frac{t^3}{n_P}\right) \tag{A.13}$$

$$= \frac{2V(P_t)}{n_P} + O\left(\frac{t^3}{n_P}\right). \tag{A.14}$$

The first-order covariance is therefore positive for small positive MI along this path: samples with a larger MI estimate also tend to have a larger variance estimate. In the two-sample statistic, P's and Q's MI estimates enter the numerator with opposite signs, whereas their variance contributions add under the square root in the standard error. At exact independence, $V(P_0) = 0$ and a second-order sampling approximation is needed.

A.4 The independence boundary

At independence, $p_{ij} = p_{i+}p_{+j}$, so every population pointwise-MI value is $\ell_P(i,j) = \log 1 = 0$. Consequently, $I(P) = V(P) = 0$, and the first-order change in MI is zero. Sampling can still produce positive estimated MI, but its leading variation comes from second-order cell errors.

To calculate this variation, fix a strictly positive independent table P and let $h = (h_{ij})$ describe a redistribution of probability, with $\sum_{i,j} h_{ij} = 0$. Define

$$P_\varepsilon = P + \varepsilon h, \qquad f(\varepsilon) = I(P_\varepsilon), \tag{A.15}$$

for ε small enough that every cell remains positive. The expansion point is $\varepsilon = 0$, where $f(0) = I(P) = 0$. The second-order Taylor formula gives

$$f(\varepsilon) = f(0) + f'(0)\varepsilon + \frac{f''(0)}{2!}\varepsilon^2 + o(\varepsilon^2). \tag{A.16}$$

The first derivative is

$$f'(0) = \sum_{i,j} \{\ell_P(i,j) - 1\} h_{ij} = -\sum_{i,j} h_{ij} = 0. \tag{A.17}$$

Here the pointwise-MI terms vanish at independence, and the remaining terms cancel because the probabilities must still sum to one.

The row and column changes are $h_{i+} = \sum_j h_{ij}$ and $h_{+j} = \sum_i h_{ij}$. Applying $\mathrm{d}^2(u \log u)/\mathrm{d}u^2 = 1/u$ to the joint and marginal terms in Equation (2.11) gives

$$f''(0) = \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}}. \tag{A.18}$$

Thus the leading change is

$$I(P_\varepsilon) = \frac{\varepsilon^2}{2} \left\{ \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}} \right\} + o(\varepsilon^2). \tag{A.19}$$

The first sum measures squared cell changes, whereas the last two remove the contributions due to changed margins, leaving the change in association. The remainder is negligible relative to ε^2 as ε approaches zero.

For a multinomial sample, take $\varepsilon h = \widehat{P} - P$. Since cell-probability errors typically have size proportional to $n^{-1/2}$, squaring them makes $\widehat{I}(P)$ proportional to n^{-1} under independence. Under fixed table dimensions and positive cell probabilities, the likelihood-ratio result is

$$2n\widehat{I}(P) \xrightarrow{d} \chi^2_{(r-1)(c-1)} \tag{A.20}$$

(Wilks, 1938; Brillinger, 2004). This is the usual one-table likelihood-ratio, or G, test of independence. Its chi-squared distribution comes from quadratic cell errors, whereas Normal Wald and Expanded Welch use the first-order MI error. The two-sample first-order approximation therefore does not apply unchanged at exact independence.

Sampling variation at small positive MI.

The quadratic approximation also explains why estimated degrees of freedom remain several units when population MI is almost zero. Keep r and c fixed and let a positive population table approach independence with cell departures proportional to $n^{-1/2}$. The normal sampling errors and the population departure then contribute on the same scale. In regular models, the likelihood-ratio statistic has a noncentral chi-squared limit under local alternatives (Banerjee, 2005). For the multinomial model

here, assume strictly positive limiting cell probabilities. The local approximation is

$$G = 2n\widehat{I}(P) \mathrel{\dot\sim} \chi^2_d(\kappa), \qquad d = (r-1)(c-1), \qquad \kappa \simeq 2nI(P), \tag{A.21}$$

where $\mathrel{\dot\sim}$ denotes an approximate distribution. The noncentral chi-squared distribution describes a sum of squared normal errors with a nonzero mean departure; κ measures that departure. At independence, $\kappa = 0$, recovering Equation (A.20).

For a sampled table close to independence, the expansions in Appendix A.2, applied around its own independent margins, give

$$\widehat{V}(P) \approx 2\widehat{I}(P) = \frac{G}{n}, \qquad \widehat{\tau}^2(P) \approx 4\widehat{V}(P) \approx \frac{4G}{n}. \tag{A.22}$$

Substitution into the implemented component formula gives

$$\widehat{\nu}_P \approx \frac{2n(G/n)^2}{4G/n} = \frac{G}{2}. \tag{A.23}$$

Thus sampling noise contributes to the estimated degrees of freedom even when the population value $nI(P)$ is very small.

The same approximation separates that plug-in calculation from matching the actual sampling moments. Since $\mathrm{E}(G) = d + \kappa$ and $\mathrm{Var}(G) = 2(d + 2\kappa)$,

$$\mathrm{E}\{\widehat{V}(P)\} \approx \frac{d+\kappa}{n}, \qquad \mathrm{Var}\{\widehat{V}(P)\} \approx \frac{2(d+2\kappa)}{n^2}. \tag{A.24}$$

Keeping both terms in the moment-matching equation gives

$$\nu_P^{\mathrm{mom}} \approx \frac{2\{(d+\kappa)/n\}^2}{2(d+2\kappa)/n^2} = \frac{(d+\kappa)^2}{d+2\kappa}. \tag{A.25}$$

When κ is much larger than d, this reduces to $\kappa/2 \simeq nI(P)$, as in Equation (4.47). When κ is small, however, the matched degrees of freedom approach d, while the mean plug-in degrees of freedom approach $d/2$.

For the 4×4, $n = 1{,}000$, $I(P) = 0.0001$ setting, $d = 9$ and $\kappa \simeq 0.2$. The local calculation gives a mean estimated component degree of freedom of approximately 4.6, rather than the population first-order value 0.1. It also predicts that the average plug-in estimate of $\mathrm{Var}\{\widehat{V}(P)\}$ divided by its actual sampling variance is approximately

$$\frac{4(d+\kappa)/n^2}{2(d+2\kappa)/n^2} = \frac{2(d+\kappa)}{d+2\kappa} \simeq 1.96. \tag{A.26}$$

This explains the scale of the estimated degrees of freedom and the roughly twofold plug-in overprediction in Table C.4. The population formula alone omits the quadratic sampling contribution responsible for both effects.

A.5 Combined degrees of freedom with a zero variance estimate

For positive component degrees of freedom, substituting Equation (4.39) and its Q analogue into Equation (4.40) gives the equivalent combined form

$$\widehat{\nu}_{\text{expanded}} = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\widehat{\tau}^2(P)/(2n_P^3) + \widehat{\tau}^2(Q)/(2n_Q^3)}. \tag{A.27}$$

This direct form remains defined when one table has $\widehat{V} = \widehat{\tau}^2 = 0$ and the other gives a positive denominator. The zero contribution drops out, leaving the other table's component degrees of freedom. The implementation nevertheless requires both components to have positive, finite degrees of freedom, so it records this case as invalid.

A.6 Numerical derivative checks

The analytic derivatives in Equations (4.18) and (4.29) were checked by centred finite differences along the one-cell perturbation paths in Chapter 4. Two positive tables,

$$\begin{pmatrix} 0.36 & 0.14 \\ 0.09 & 0.41 \end{pmatrix}, \qquad \begin{pmatrix} 0.28 & 0.09 & 0.07 \\ 0.08 & 0.17 & 0.06 \\ 0.04 & 0.10 & 0.11 \end{pmatrix},$$

were checked at every cell with step size $\varepsilon = 10^{-6}$. For each path, the difference between the values at $+\varepsilon$ and $-\varepsilon$ was divided by 2ε; all perturbed probabilities remained positive. The maximum absolute error was of order 10^{-10} for both derivatives.

Appendix B

Experimental Details

This appendix specifies the population formulas, target-MI search, complete parameter grids, sampling procedure and outcome calculations. Together with the test formulas in Chapters 3 and 4, these details define the statistical experiment.

B.1 Copying construction and target selection

For a positive marginal vector a, the joint distribution is

$$p_{ij}(\lambda) = (1-\lambda)a_i a_j + \lambda a_i \mathbf{1}_{\{i=j\}}. \tag{B.1}$$

Summing a row gives

$$\sum_j p_{ij}(\lambda) = (1-\lambda)a_i \sum_j a_j + \lambda a_i = a_i. \tag{B.2}$$

Summing a column similarly gives a_j. The copying branch contributes only to its matching category, while the independent branch contributes across every category. Thus the construction preserves both margins and has positive cells whenever $0 \leq \lambda < 1$.

MI increases continuously from zero to the entropy of a. With the margins fixed, differentiating the joint terms gives

$$\frac{\mathrm{d}^2 I\{P(\lambda)\}}{\mathrm{d}\lambda^2} = \sum_{i,j} \frac{(a_i \mathbf{1}_{\{i=j\}} - a_i a_j)^2}{p_{ij}(\lambda)} > 0. \tag{B.3}$$

The first derivative at zero is zero, because the change in probabilities sums to zero and all pointwise MI values are zero there. The positive second derivative makes

the first derivative positive for $\lambda > 0$. At the endpoint, $Y = X$, so

$$I\{P(1)\} = -\sum_i a_i \log a_i. \tag{B.4}$$

These facts give a unique copying probability for each positive target below the entropy endpoint.

For each population, calculate this endpoint and solve $I\{P(\lambda)\} - I_{\text{target}} = 0$ on $[0, 1]$ using Brent's method (Brent, 1973). Absolute and relative root tolerances are 10^{-15} and 10^{-14}. The endpoint brackets the search; sampled populations use the interior solution. Accept a table only when every cell is positive, its total and marginal errors are at most 10^{-12}, and its MI error is at most $10^{-12} + 10^{-8} I_{\text{target}}$. The maximum target in the study is 0.22 nats, below the smallest chosen entropy endpoint, approximately 0.5004 nats.

B.2 Four-cell supporting construction

For the supporting 4×4 and 16×16 comparisons, begin with the independence probabilities $a_i a_j$ and transfer an amount t within the first two rows and columns. The resulting table $R(t)$ has cell probabilities

$$r_{ij}(t) = \begin{cases} a_i a_j + t, & (i, j) = (1, 1) \text{ or } (2, 2), \\ a_i a_j - t, & (i, j) = (1, 2) \text{ or } (2, 1), \\ a_i a_j, & \text{all other cells.} \end{cases} \tag{B.5}$$

Every affected row and column gains and loses t, preserving its total. The decreasing cells have starting probability $a_1 a_2$, so

$$0 \le t < t_{\max} = a_1 a_2. \tag{B.6}$$

Solve $I\{R(t)\} = I_{\text{target}}$ by the same root search on $[0, t_{\max}]$. The target must lie below the endpoint MI, and the constructed table passes the same checks as the copying table.

In a binary table the copying construction is exactly this transfer:

$$P(\lambda) = (1 - \lambda) \begin{pmatrix} a_1^2 & a_1 a_2 \\ a_1 a_2 & a_2^2 \end{pmatrix} + \lambda \begin{pmatrix} a_1 & 0 \\ 0 & a_2 \end{pmatrix} \tag{B.7}$$

$$= \begin{pmatrix} a_1^2 & a_1 a_2 \\ a_1 a_2 & a_2^2 \end{pmatrix} + \lambda a_1 a_2 \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \tag{B.8}$$

The second equality uses $a_1 + a_2 = 1$, giving $a_1 - a_1^2 = a_1 a_2$ and $a_2 - a_2^2 = a_1 a_2$. Hence $t = \lambda a_1 a_2$. This equivalence does not extend to the larger tables: copying changes cells throughout the table, whereas Equation (B.5) changes only four.

B.3 Complete parameter grids and sampling

Table B.1 lists the parameter combinations for each study and supporting check. Table sizes are $k \in \{2, 4, 8, 16\}$. Each population has identical row and column margins. Uniform margins give probability $1/k$ to every category; same skew uses $m_k(0.8)$ in both populations; different skew uses $m_k(0.7)$ in P and $m_k(0.8)$ in Q, with m_k defined by Equation (5.3). The same copying formula and root search apply to every primary-study setting.

Table B.1: Exact parameter sets; MI values are in nats.

Study	Settings
Equal MI	All four sizes and three margin regimes; $I(P) = I(Q) = 0.02$; equal sample sizes $n \in \{10, 20, 50, 100, 250, 500, 1000, 2500, 10000, 50000\}$.
Population MI	4×4, different skew; $I(P) = I(Q) \in \{0.0001, 0.0003, 0.001, 0.003, 0.01, 0.02, 0.05\}$; $n \in \{100, 1000\}$.
Power	All four sizes with different skew, plus uniform and same-skew 4×4; $I(P) = 0.02$, $I(Q) = 0.02 + \lvert\Delta\rvert$; $\lvert\Delta\rvert \in \{0, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2\}$; $n \in \{1000, 10000\}$.
Four-cell check	$k \in \{4, 16\}$, different skew; $(I(P), I(Q)) \in \{(0.02, 0.02), (0.02, 0.03)\}$; $n \in \{1000, 10000\}$; construction in Equation (B.5).
Sample allocation	Copying 4×4, different skew; $(I(P), I(Q)) \in \{(0.02, 0.02), (0.02, 0.03)\}$; $(n_P, n_Q) \in \{(1000, 1000), (400, 1600), (1600, 400)\}$. The total sample size is 2,000 in every comparison.

For one repetition, draw

$$(N_{P,11}, \ldots, N_{P,kk}) \sim \text{Multinomial}(n_P; p_{11}, \ldots, p_{kk}), \tag{B.9}$$

and independently draw the corresponding Q counts. Reshape these counts into their $k \times k$ tables and apply both methods to that pair. Repeat independently 20,000 times per distinct setting. The simulations use the PCG64 pseudo-random generator with master seed 20261006 and separate streams for the populations and settings. Repeated settings use one set of repetitions in every relevant display.

All samples retain their declared table dimensions, including empty rows and columns. Use natural logarithms, $0 \log 0 = 0$, and no smoothing. The bias correction for each plug-in MI is $(k-1)^2/(2n)$; do not truncate negative corrected estimates. In the equal-sample studies, the two corrections cancel in their difference. In the allocation check, Equation (3.7) subtracts $9/(2n_P)$ and adds $9/(2n_Q)$ to the plug-in difference. Their combined contribution is -0.0084375 nats at $(400, 1600)$ and $+0.0084375$ at $(1600, 400)$. These terms remove the leading unequal-sample bias; the diagnostics measure the remaining bias. Calculate the standard error using Equation (3.9) and the component and combined degrees of freedom using Chapter 4.

B.4 Validity and sparse-sample measures

Normal Wald requires a finite corrected difference, a finite positive standard error, $\widehat{V}(P)+\widehat{V}(Q) > 10^{-14}$, and a finite p-value. The threshold treats variances extremely close to zero as numerically unusable. Expanded Welch additionally requires positive finite degrees of freedom for both components and their combination, and a finite expanded p-value. A failed component makes Expanded Welch invalid under this rule, including the zero-component case in Appendix A.5.

The primary rate divides valid rejections by all repetitions. A method-valid rate uses only that method's valid repetitions; a common-valid rate uses only pairs where both methods are valid. Report the denominator for either conditional rate, which is undefined when the denominator is zero. Invalid outputs contribute no rejection to the primary rate.

Record the number of zero-count cells, empty rows and empty columns separately for each population. These are sampling diagnostics rather than eligibility rules. The minimum expected cell count is $\min_{ij} np_{ij}$, with the corresponding definition for Q. The study applies no minimum expected-count filter.

B.5 Bias, variance and degrees-of-freedom diagnostics

Let $\widehat{\Delta}_s$ be the corrected difference in repetition s. Its bias estimate is the mean of the finite-valued MI difference estimates minus the known Δ. Its observed sampling variance is

$$\frac{1}{M-1}\sum_{s=1}^{M}\left(\widehat{\Delta}_s - \frac{1}{M}\sum_{u=1}^{M}\widehat{\Delta}_u\right)^2, \tag{B.10}$$

where M counts those finite-valued estimates. Centre at their sample mean, rather than the true difference, so this measures variance separately from bias. Include finite-valued MI differences even when the test output is invalid. For P and Q, the individual corrected MI biases are the mean finite corrected MI estimates minus $I(P)$ and $I(Q)$, respectively.

For the estimated/observed variance ratio, restrict both averages to the same repetitions with a finite difference and finite squared standard error. Retain finite zero variance estimates. Divide the mean squared standard error by the observed variance on this matched subset; the ratio is undefined if the observed variance is zero. Report its included fraction. The population comparison instead uses $V(P)/n_P + V(Q)/n_Q$ and the primary observed variance.

The Monte Carlo standard error of estimated bias is the observed SD divided by $\sqrt{M}$. Degrees-of-freedom medians and percentiles use valid Expanded Welch repetitions, with their included fraction reported. Percentile ranges describe the spread of those outputs, not confidence intervals for their median. These summaries expose estimation problems but do not isolate the causal contribution of each problem to rejection.

Variability of the estimated MI variance.

For each selected population, measure the observed variability of $\widehat{V}(P)$ by replacing $\widehat{\Delta}_s$ with $\widehat{V}_s(P)$ in Equation (B.10). This measures how much the variance estimate itself changes between samples. Chapter 4 predicts that variability to first order. Compare the observed value with

$$\begin{aligned}\text{population prediction} &= \frac{\tau^2(P)}{n_P},\\ \text{mean plug-in prediction} &= \frac{1}{M}\sum_{s=1}^{M}\frac{\widehat{\tau}_s^2(P)}{n_P}.\end{aligned} \tag{B.11}$$

The population prediction uses the known probabilities, whereas the mean plug-in prediction uses the empirical probabilities in each repetition. Repeat both for Q.

For each population, use the same repetitions with finite $\widehat{V}_s$ and $\widehat{\tau}_s^2$, including finite zeros and outputs whose test is invalid. Divide each prediction by the observed variability: a ratio of one means agreement, and the ratio is undefined when observed variability is zero.

Estimated direction among rejections.

For a nonzero population difference, classify each valid two-sided rejection by the sign of $\widehat{\Delta}_{\mathrm{BC}}$. The estimate has the opposite sign when $\widehat{\Delta}_{\mathrm{BC}}\Delta < 0$; it has the correct sign when $\widehat{\Delta}_{\mathrm{BC}}\Delta > 0$. Report the number of rejections and the number with opposite signs for each method. The fraction with opposite signs divides the opposite-sign count by the rejection count and is undefined if there are no rejections. These counts supplement the estimate of power, which includes every valid rejection regardless of direction.

B.6 Simulation uncertainty

For a rate estimate $\widehat{r}$ from M independent repetitions, its approximate Monte Carlo standard error is $\{\widehat{r}(1-\widehat{r})/M\}^{1/2}$. Wilson intervals use $z = \Phi^{-1}(0.975)$ and have centre and half-width

$$\frac{\widehat{r} + z^2/(2M)}{1 + z^2/M}, \qquad \frac{z\sqrt{\widehat{r}(1-\widehat{r})/M + z^2/(4M^2)}}{1 + z^2/M}. \tag{B.12}$$

These intervals are used for both rejection and validity rates (Wilson, 1927).

For the paired method contrast, let D_s equal the Wald valid-rejection indicator minus the Expanded Welch valid-rejection indicator on pair s. The approximate interval for the average paired difference $\bar{D}$ is

$$\bar{D} \;\pm\; 1.96 \left\{ \frac{1}{M(M-1)} \sum_{s=1}^{M} (D_s - \bar{D})^2 \right\}^{1/2}. \tag{B.13}$$

Use the same common-valid subset when reporting a conditional paired contrast. Cross-construction comparisons use independent simulation streams, as do different sample allocations, so their difference uncertainty is not calculated from paired indicators. All intervals describe simulation uncertainty at fixed populations, not simultaneous coverage over every setting.

B.7 Timing procedure

Scalar-call timings are measured on an arm64 machine running macOS 26.6.2, Python 3.13.5, NumPy 2.4.4 and SciPy 1.17.1. Each setting uses 200 pre-generated different-skew null table pairs at $n = 1{,}000$, covering 2×2 through 16×16 tables. Sampling and population construction are excluded from the timings. After 20 warm-up calls per method, five timed passes are made through the 200 pairs. Calls are timed individually, and the method order alternates between passes. The median and interquartile range of the resulting 1,000 calls per method and setting are reported in Table 6.2.

Appendix C

Additional Results

This appendix gives the complete simulation results and supporting checks for Chapter 6, followed by Holm's multiple-testing adjustment for the online-shopping application. Each simulation setting uses 20,000 sampled table pairs and the two-sided 5% threshold.

C.1 Complete equal-MI results

Table C.1 reports every combination of four table sizes, three margin regimes and ten sample sizes from the equal-MI study. Both population MI values are 0.02 nats. Rejection rates divide valid rejections by all repetitions, while the final two columns report valid-result frequencies for Wald and Expanded Welch. The margin vectors are those in Table 5.2.

Table C.1: False-positive and valid-result rates for all 120 null settings.

Size	Margins	n	Wald	Expanded	Valid W	Valid E
2×2	Uniform	10	0.0591	0.0185	0.9907	0.8063
2×2	Uniform	20	0.0276	0.0169	0.9985	0.9260
2×2	Uniform	50	0.0081	0.0042	0.9999	0.9785
2×2	Uniform	100	0.0128	0.0054	1.0000	0.9968
2×2	Uniform	250	0.0302	0.0179	1.0000	1.0000
2×2	Uniform	500	0.0387	0.0276	1.0000	1.0000
2×2	Uniform	1,000	0.0446	0.0390	1.0000	1.0000
2×2	Uniform	2,500	0.0490	0.0464	1.0000	1.0000
2×2	Uniform	10,000	0.0465	0.0458	1.0000	1.0000
2×2	Uniform	50,000	0.0491	0.0490	1.0000	1.0000

Size	Margins	n	Wald	Expanded	Valid W	Valid E
2×2	Same skew	10	0.0358	0.0092	0.9572	0.6338
2×2	Same skew	20	0.0239	0.0136	0.9976	0.9083
2×2	Same skew	50	0.0095	0.0046	1.0000	0.9890
2×2	Same skew	100	0.0114	0.0040	1.0000	0.9971
2×2	Same skew	250	0.0264	0.0133	1.0000	1.0000
2×2	Same skew	500	0.0390	0.0255	1.0000	1.0000
2×2	Same skew	1,000	0.0452	0.0372	1.0000	1.0000
2×2	Same skew	2,500	0.0483	0.0456	1.0000	1.0000
2×2	Same skew	10,000	0.0500	0.0490	1.0000	1.0000
2×2	Same skew	50,000	0.0495	0.0494	1.0000	1.0000
2×2	Different skew	10	0.0407	0.0109	0.9816	0.7176
2×2	Different skew	20	0.0271	0.0150	0.9981	0.9167
2×2	Different skew	50	0.0087	0.0045	1.0000	0.9912
2×2	Different skew	100	0.0107	0.0048	1.0000	0.9974
2×2	Different skew	250	0.0274	0.0137	1.0000	1.0000
2×2	Different skew	500	0.0404	0.0278	1.0000	1.0000
2×2	Different skew	1,000	0.0425	0.0360	1.0000	1.0000
2×2	Different skew	2,500	0.0497	0.0466	1.0000	1.0000
2×2	Different skew	10,000	0.0504	0.0496	1.0000	1.0000
2×2	Different skew	50,000	0.0495	0.0494	1.0000	1.0000
4×4	Uniform	10	0.0985	0.0511	1.0000	0.9992
4×4	Uniform	20	0.0813	0.0470	1.0000	1.0000
4×4	Uniform	50	0.0472	0.0309	1.0000	1.0000
4×4	Uniform	100	0.0210	0.0144	1.0000	1.0000
4×4	Uniform	250	0.0217	0.0154	1.0000	1.0000
4×4	Uniform	500	0.0283	0.0218	1.0000	1.0000
4×4	Uniform	1,000	0.0383	0.0331	1.0000	1.0000
4×4	Uniform	2,500	0.0442	0.0420	1.0000	1.0000
4×4	Uniform	10,000	0.0467	0.0458	1.0000	1.0000
4×4	Uniform	50,000	0.0490	0.0490	1.0000	1.0000
4×4	Same skew	10	0.0798	0.0271	0.9606	0.6401
4×4	Same skew	20	0.0479	0.0308	0.9996	0.9534
4×4	Same skew	50	0.0142	0.0075	1.0000	0.9999

Size	Margins	n	Wald	Expanded	Valid W	Valid E
4×4	Same skew	100	0.0068	0.0037	1.0000	1.0000
4×4	Same skew	250	0.0192	0.0085	1.0000	1.0000
4×4	Same skew	500	0.0325	0.0208	1.0000	1.0000
4×4	Same skew	1,000	0.0392	0.0319	1.0000	1.0000
4×4	Same skew	2,500	0.0438	0.0400	1.0000	1.0000
4×4	Same skew	10,000	0.0481	0.0472	1.0000	1.0000
4×4	Same skew	50,000	0.0508	0.0506	1.0000	1.0000
4×4	Different skew	10	0.1332	0.0465	0.9910	0.7588
4×4	Different skew	20	0.0801	0.0443	1.0000	0.9772
4×4	Different skew	50	0.0240	0.0118	1.0000	0.9999
4×4	Different skew	100	0.0124	0.0059	1.0000	1.0000
4×4	Different skew	250	0.0238	0.0119	1.0000	1.0000
4×4	Different skew	500	0.0332	0.0225	1.0000	1.0000
4×4	Different skew	1,000	0.0394	0.0331	1.0000	1.0000
4×4	Different skew	2,500	0.0427	0.0399	1.0000	1.0000
4×4	Different skew	10,000	0.0481	0.0473	1.0000	1.0000
4×4	Different skew	50,000	0.0476	0.0472	1.0000	1.0000
8×8	Uniform	10	0.1397	0.0800	1.0000	0.9996
8×8	Uniform	20	0.0703	0.0584	1.0000	1.0000
8×8	Uniform	50	0.0689	0.0586	1.0000	1.0000
8×8	Uniform	100	0.0737	0.0651	1.0000	1.0000
8×8	Uniform	250	0.0362	0.0329	1.0000	1.0000
8×8	Uniform	500	0.0222	0.0207	1.0000	1.0000
8×8	Uniform	1,000	0.0266	0.0245	1.0000	1.0000
8×8	Uniform	2,500	0.0333	0.0318	1.0000	1.0000
8×8	Uniform	10,000	0.0431	0.0426	1.0000	1.0000
8×8	Uniform	50,000	0.0498	0.0495	1.0000	1.0000
8×8	Same skew	10	0.1054	0.0488	0.9606	0.6413
8×8	Same skew	20	0.0942	0.0738	0.9994	0.9529
8×8	Same skew	50	0.0503	0.0361	1.0000	1.0000
8×8	Same skew	100	0.0189	0.0121	1.0000	1.0000
8×8	Same skew	250	0.0067	0.0045	1.0000	1.0000
8×8	Same skew	500	0.0084	0.0060	1.0000	1.0000

Size	Margins	n	Wald	Expanded	Valid W	Valid E
8×8	Same skew	1,000	0.0242	0.0199	1.0000	1.0000
8×8	Same skew	2,500	0.0414	0.0388	1.0000	1.0000
8×8	Same skew	10,000	0.0456	0.0444	1.0000	1.0000
8×8	Same skew	50,000	0.0503	0.0499	1.0000	1.0000
8×8	Different skew	10	0.1939	0.0991	0.9890	0.7605
8×8	Different skew	20	0.1805	0.1432	1.0000	0.9762
8×8	Different skew	50	0.1218	0.0957	1.0000	1.0000
8×8	Different skew	100	0.0706	0.0518	1.0000	1.0000
8×8	Different skew	250	0.0384	0.0289	1.0000	1.0000
8×8	Different skew	500	0.0276	0.0217	1.0000	1.0000
8×8	Different skew	1,000	0.0311	0.0268	1.0000	1.0000
8×8	Different skew	2,500	0.0412	0.0384	1.0000	1.0000
8×8	Different skew	10,000	0.0467	0.0454	1.0000	1.0000
8×8	Different skew	50,000	0.0487	0.0483	1.0000	1.0000
16×16	Uniform	10	0.1764	0.0455	1.0000	0.9960
16×16	Uniform	20	0.1051	0.0822	1.0000	1.0000
16×16	Uniform	50	0.0460	0.0425	1.0000	1.0000
16×16	Uniform	100	0.0523	0.0485	1.0000	1.0000
16×16	Uniform	250	0.0859	0.0812	1.0000	1.0000
16×16	Uniform	500	0.0655	0.0635	1.0000	1.0000
16×16	Uniform	1,000	0.0339	0.0331	1.0000	1.0000
16×16	Uniform	2,500	0.0228	0.0222	1.0000	1.0000
16×16	Uniform	10,000	0.0348	0.0344	1.0000	1.0000
16×16	Uniform	50,000	0.0469	0.0469	1.0000	1.0000
16×16	Same skew	10	0.1099	0.0593	0.9580	0.6347
16×16	Same skew	20	0.1179	0.0982	0.9995	0.9527
16×16	Same skew	50	0.0888	0.0732	1.0000	1.0000
16×16	Same skew	100	0.0537	0.0443	1.0000	1.0000
16×16	Same skew	250	0.0190	0.0155	1.0000	1.0000
16×16	Same skew	500	0.0087	0.0074	1.0000	1.0000
16×16	Same skew	1,000	0.0065	0.0059	1.0000	1.0000
16×16	Same skew	2,500	0.0144	0.0134	1.0000	1.0000
16×16	Same skew	10,000	0.0450	0.0440	1.0000	1.0000

Size	Margins	n	Wald	Expanded	Valid W	Valid E
16×16	Same skew	50,000	0.0464	0.0462	1.0000	1.0000
16×16	Different skew	10	0.2063	0.1190	0.9900	0.7571
16×16	Different skew	20	0.2301	0.2003	1.0000	0.9750
16×16	Different skew	50	0.2201	0.2016	1.0000	1.0000
16×16	Different skew	100	0.2155	0.1984	1.0000	1.0000
16×16	Different skew	250	0.2352	0.2181	1.0000	1.0000
16×16	Different skew	500	0.2700	0.2551	1.0000	1.0000
16×16	Different skew	1,000	0.2522	0.2402	1.0000	1.0000
16×16	Different skew	2,500	0.0925	0.0882	1.0000	1.0000
16×16	Different skew	10,000	0.0463	0.0453	1.0000	1.0000
16×16	Different skew	50,000	0.0477	0.0474	1.0000	1.0000

Figure C.1 supplies the four size–margin curves not displayed in the main null figures: uniform and same-skew 2×2 and 8×8 populations. Error bars are pointwise Wilson 95% intervals; the lower strips show validity.

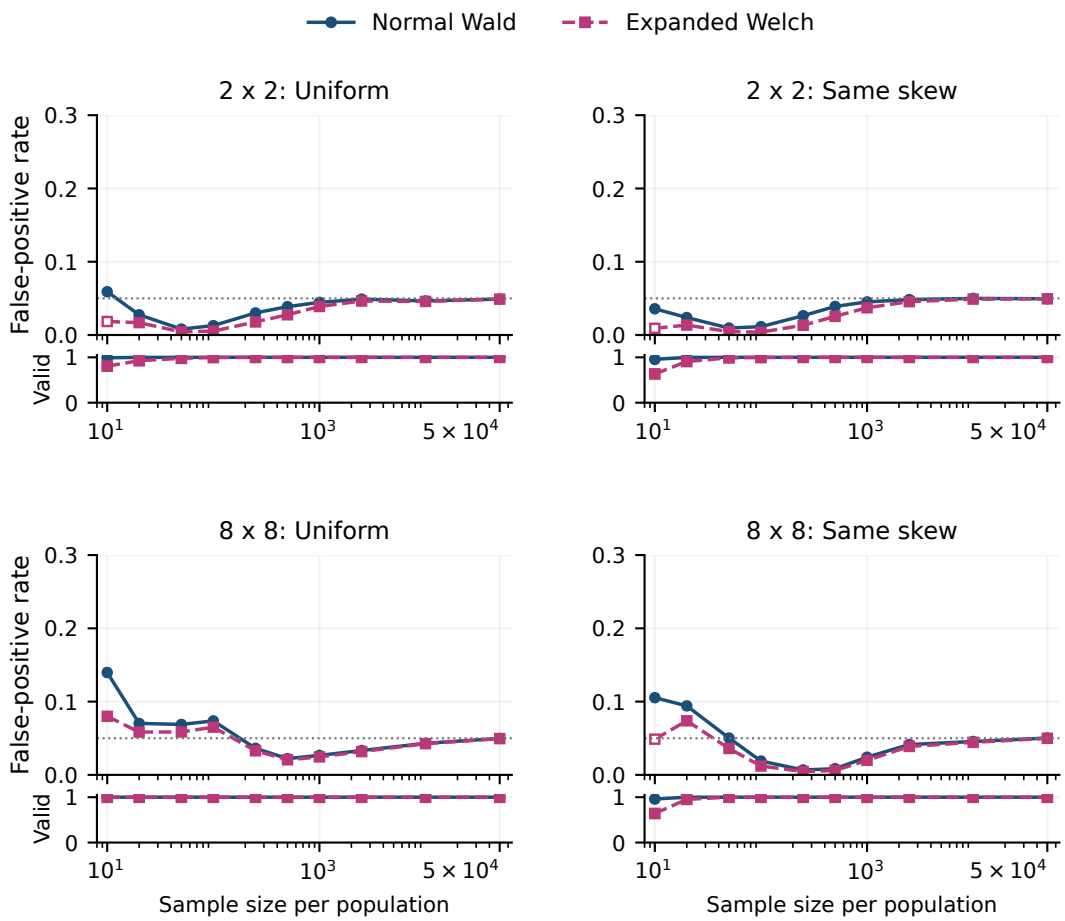


Figure C.1: Additional equal-MI comparisons under uniform and same-skew margins.

The smaller tables again give low false-positive rates at intermediate sample sizes. At $n = 100$, Wald rejects at 0.0128 under uniform margins and 0.0114 under same skew, with still lower Expanded Welch rates. In 8×8 tables at $n = 100$, uniform margins give rates 0.0737 and 0.0651 for Wald and Expanded Welch, respectively; same-skew margins give 0.0189 and 0.0121. The effect of the margin regime therefore changes with table size. By $n = 50{,}000$, all four comparisons have rates between 0.0490 and 0.0503 and full validity.

C.2 Supporting estimation diagnostics

Table C.2 gives the diagnostics behind the population-MI comparison in Figure 6.3. Est./obs. is the ratio of mean squared standard error to observed sampling variance. Pop./obs. replaces the estimated variance with the population first-order variance $V(P)/n + V(Q)/n$. DF denotes combined degrees of freedom; its bracketed range gives the 5th and 95th percentiles. All repetitions have finite diagnostics and valid test outputs in these settings.

Table C.2: Diagnostics for 4×4 different-skew populations with equal MI.

n	MI	Bias	DF median [5th, 95th]	Est./obs.	Pop./obs.
100	0.0001	0.00751	4.23 [2.25, 8.77]	1.919	0.007
100	0.0003	0.00724	4.33 [2.27, 9.11]	1.850	0.019
100	0.001	0.00666	4.53 [2.32, 9.62]	1.851	0.062
100	0.003	0.00606	4.98 [2.45, 10.63]	1.760	0.166
100	0.01	0.00567	6.35 [2.92, 13.61]	1.611	0.416
100	0.02	0.00569	8.10 [3.65, 17.17]	1.432	0.608
100	0.05	0.00588	13.80 [6.32, 27.97]	1.176	0.808
1,000	0.0001	-0.00011	8.62 [4.75, 14.47]	1.736	0.040
1,000	0.0003	-0.00018	8.79 [4.86, 14.52]	1.733	0.114
1,000	0.001	-0.00017	9.62 [5.44, 15.50]	1.590	0.305
1,000	0.003	-0.00015	12.32 [7.21, 19.21]	1.377	0.579
1,000	0.01	-0.00022	21.93 [14.18, 31.82]	1.163	0.835
1,000	0.02	-0.00012	35.70 [24.60, 49.15]	1.061	0.895
1,000	0.05	-0.00003	78.24 [59.53, 100.65]	1.004	0.948

At the smallest MI targets, the population first-order variance underestimates observed sampling variance, while the mean squared standard error overestimates it. Both ratios move closer to one as MI increases, especially at $n = 1{,}000$. An overestimated standard error reduces the statistic's magnitude, and Expanded Welch's larger rejection cutoff makes rejection still less likely.

Table C.3 continues the main diagnostic comparison to $n = 50{,}000$ at common MI 0.02. All outputs are valid. At 4×4 the difference bias has Monte Carlo standard error approximately 0.000010 nats; at 16×16 it is approximately 0.000012. The biases are small, the estimated/observed variance ratios are near one, and the large degrees of freedom bring the Student t distribution close to the standard normal distribution. The population first-order variance still differs more from observed sampling variance at 16×16 than at 4×4.

Table C.3: Different-skew null diagnostics at $n = 50{,}000$.

Size	Bias	Est./obs.	Pop./obs.	DF median
4×4	-0.00002	1.006	1.002	1436.59
16×16	-0.00002	1.017	0.951	1146.78

Variability of the estimated MI variance.

Table C.4 compares predicted and observed sampling variability of $\widehat{V}(P)$ and $\widehat{V}(Q)$, used to choose Expanded Welch's degrees of freedom. Observed variability is the sample variance across repetitions. For P, Pop./obs. divides $\tau^2(P)/n_P$ by this observed value; plug-in/obs. uses the average $\widehat{\tau}^2(P)/n_P$, with corresponding definitions for Q. The predictions are defined in Equation (B.11). The six selected settings use different-skew copying populations, and each row includes all 20,000 finite outputs.

Table C.4: Predicted and observed variability of the estimated MI variance.

Size	n	Common MI	Population	Pop./obs.	Plug-in/obs.
4×4	1,000	0.0001	P	0.047	1.950
4×4	1,000	0.0001	Q	0.051	2.093
4×4	1,000	0.02	P	0.929	1.039
4×4	1,000	0.02	Q	0.985	1.097
4×4	1,000	0.05	P	1.023	1.028
4×4	1,000	0.05	Q	1.015	1.009
4×4	10,000	0.02	P	0.993	1.004
4×4	10,000	0.02	Q	1.007	1.017
16×16	1,000	0.02	P	0.768	1.696
16×16	1,000	0.02	Q	0.720	1.824
16×16	10,000	0.02	P	0.980	1.094
16×16	10,000	0.02	Q	1.008	1.106

At 4×4, common MI 0.02 and $n = 10{,}000$, both predictions are close to observed variability. At common MI 0.05 and $n = 1{,}000$, the ratios are also close to one. Near independence at common MI 0.0001, however, the population first-order prediction accounts for about 5% of observed variability, while the plug-in prediction is approximately twice that variability. Thus the approximation at the true table and substitution of empirical probabilities produce different errors.

The larger table gives another intermediate case. At 16×16 and $n = 1{,}000$, the population predictions account for approximately 72–77% of observed variability, while the plug-in predictions are 1.70–1.82 times that variability. At $n = 10{,}000$, the population ratios are close to one, but the plug-in predictions remain about 10% too large. These comparisons assess the sampling approximation used to calculate

degrees of freedom. The usefulness of the resulting cutoff also depends on the shared MI statistic, as the main results demonstrate.

C.3 A jackknife check of MI difference bias

This check asks whether changing the bias correction improves the sparse MI differences that give wrong-direction rejections in the main results. The delete-one jackknife uses the change in an estimate after removing one observation to estimate its leading finite-sample bias (Paninski, 2003, Section 3).

For the sample from P, let $\widehat{I}^{(-a)}(P)$ be the raw plug-in MI after removing observation a. Each such estimate uses the remaining $n_P - 1$ observations and recalculates both margins. Subtracting $n_P - 1$ times the change between the average leave-one-out estimate and the full-sample estimate gives

$$\begin{aligned}\widehat{I}_{\mathrm{J}}(P) &= \widehat{I}(P) - (n_P - 1)\left\{\frac{1}{n_P}\sum_{a=1}^{n_P}\widehat{I}^{(-a)}(P) - \widehat{I}(P)\right\} \\ &= n_P\widehat{I}(P) - \frac{n_P - 1}{n_P}\sum_{a=1}^{n_P}\widehat{I}^{(-a)}(P). \end{aligned} \tag{C.1}$$

Because observations in the same cell give identical leave-one-out tables, the average can be calculated once per occupied cell, weighting its leave-one-out MI by N_{ij}^P/n_P. A cell reduced to zero contributes zero to the MI sum. Apply the same calculation to Q, retaining all row and column categories and any negative corrected estimates.

The jackknife replaces the analytic correction in Equation (3.7). The estimated difference and statistic become

$$\widehat{\Delta}_{\mathrm{J}} = \widehat{I}_{\mathrm{J}}(P) - \widehat{I}_{\mathrm{J}}(Q), \qquad T_{\mathrm{J}} = \frac{\widehat{\Delta}_{\mathrm{J}}}{\sqrt{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q}}. \tag{C.2}$$

The variance estimates in the denominator are unchanged. Compare $|T_{\mathrm{J}}|$ with the 97.5th percentile of the standard normal distribution, approximately 1.96, giving the same two-sided 5% cutoff as Wald. This separates changing the MI estimate from changing its estimated uncertainty.

Use 16×16 copying populations. In both row and column margins, the first category has probability 0.7 for P and 0.8 for Q. Each remaining category has probability 0.3/15 or 0.2/15, respectively. Copying sets $Y = X$ with a probability chosen to attain the target MI; otherwise, Y is drawn independently with the same margins as X. Fix $I(P) = 0.02$ and use $I(Q) = 0.02$ or 0.03 nats, each at $n_P = n_Q = 1{,}000$ and 10,000. The two corrections use the same 20,000 table pairs at

each setting. All calculations are valid, and rejection rates divide by all repetitions. In the table, Bias is mean estimation error in nats, Obs. SD is observed standard deviation, and Est./obs. is the ratio of mean squared standard error to observed variance.

Table C.5: Analytic and jackknife MI differences with the same standard error.

n	Δ	Correction	Bias	Obs. SD	Est./obs.	Rejection rate
1,000	0.00	Analytic	+0.02960	0.01449	1.94	0.2522
1,000	0.00	Jackknife	-0.00116	0.01521	1.76	0.0085
1,000	-0.01	Analytic	+0.03061	0.01564	1.81	0.1005
1,000	-0.01	Jackknife	-0.00024	0.01638	1.65	0.0275
10,000	0.00	Analytic	-0.00088	0.00387	1.07	0.0463
10,000	0.00	Jackknife	-0.00045	0.00401	1.00	0.0507
10,000	-0.01	Analytic	-0.00094	0.00433	1.05	0.6992
10,000	-0.01	Jackknife	-0.00057	0.00447	0.99	0.6619

Table C.5 shows a much smaller difference bias at $n = 1{,}000$. Under equal MI, it falls from 0.02960 to -0.00116 nats; under the alternative, from 0.03061 to -0.00024 nats. The individual jackknife estimates remain upward biased. Their null biases are 0.03439 nats for P and 0.03555 for Q. Under the alternative, the jackknife MI biases are 0.03452 for P and 0.03476 nats for Q. These similar biases largely cancel in the difference. The observed standard deviation increases slightly, but the root mean squared error falls from 0.03295 to 0.01525 nats under the null and from 0.03438 to 0.01638 under the alternative.

The jackknife difference also gives a more accurate estimated ordering at the sparse alternative. Opposite-direction rejections fall from 2,009 to 5, while correct-direction rejections rise from 1 to 544. The total rejection rate falls because the original rejections almost all indicate the wrong ordering. Null rejection, however, falls to 0.0085, with a Wilson 95% Monte Carlo interval of [0.0073, 0.0098]. The mean squared standard error still exceeds the observed variance of the jackknife difference: their ratio is 1.76 under the null and 1.65 under the alternative. Better estimation of the difference therefore leaves a separate problem in estimating its uncertainty.

At $n = 10{,}000$, the jackknife null rejection rate is 0.0507, with a Wilson interval of [0.0477, 0.0538]. The estimated/observed variance ratios for the analytic and

jackknife differences are both close to one. Alternative rejection falls from 0.6992 to 0.6619, with every rejection indicating the correct direction for both corrections. The jackknife difference has less bias but slightly greater observed variability and root mean squared error at this sample size. This check supports its use for improving the identified sparse differences, while showing that the gain does not extend to every performance measure or sample size.

C.4 Power over the full MI-difference range

Figures C.2 and C.3 show the same settings as the main power comparisons, extending the horizontal range through 0.2 nats. They retain the zero-difference points and Wilson interval bands. The population targets are $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$; every result is valid in these settings.

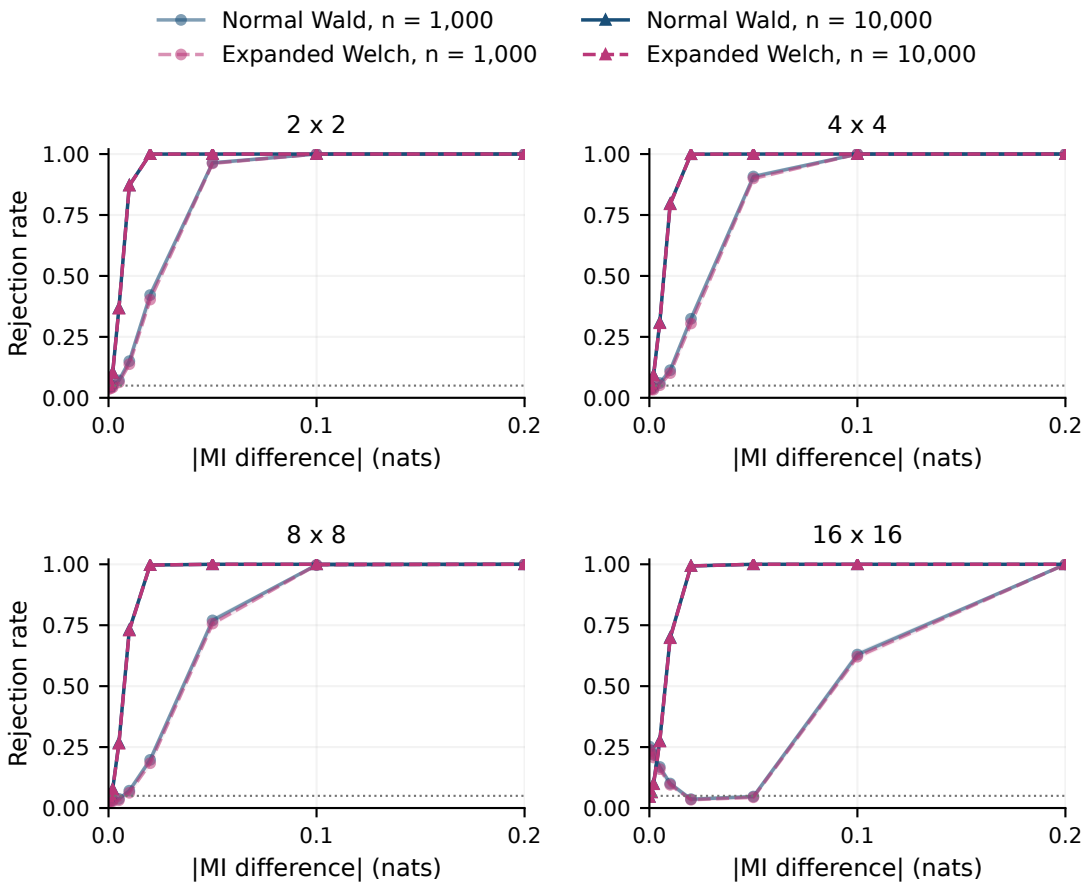


Figure C.2: Full-range rejection rates under different-skew margins.

At $n = 1{,}000$, power for a 0.05-nat difference falls with table size: Wald rates are 0.9642, 0.9079, 0.7699 and 0.0471 from 2×2 through 16×16. The last curve

first falls from its high null rate, then rises to 0.6297 at 0.1 nats and 0.9997 at 0.2. Unequal bias opposes the small negative true differences there, as discussed in the main results. At $n = 10{,}000$, both methods detect every simulated 0.05-nat difference at all four sizes.

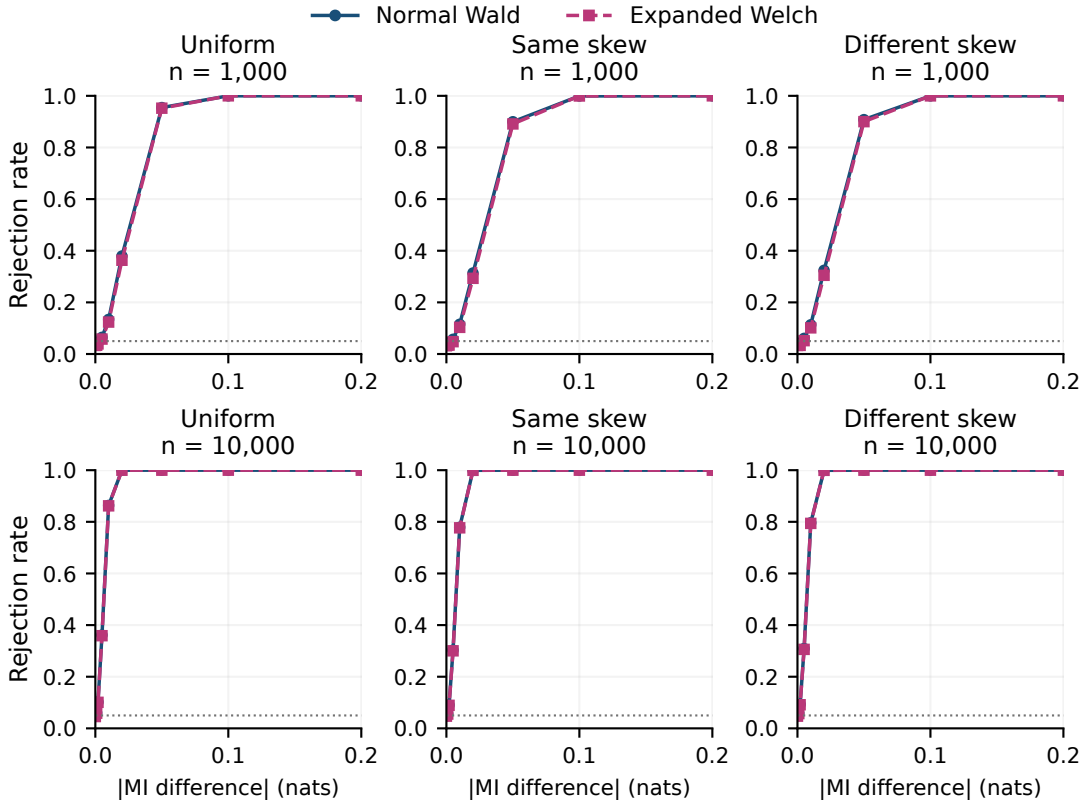


Figure C.3: Full-range rejection rates across 4×4 margin regimes.

The margin regimes differ most at intermediate effects, before rejection approaches one. At 0.2 nats, both methods reject every simulated sample at both sample sizes in all three regimes. Near this upper limit, a smaller difference between methods says little about their relative performance for weaker effects; the enlarged main figures show those effects more clearly.

Estimated direction among rejections.

Table C.6 examines the 16×16 different-skew copying populations at two small MI differences. $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$, so the true signed difference is negative. Opposite-sign rejections have $\widehat{\Delta}_{\mathrm{BC}} > 0$, indicating larger estimated MI for P. Every setting has 20,000 repetitions and full validity; the opposite-sign count is a subset of the rejection count, not a percentage of all samples.

Table C.6: Estimated direction among two-sided rejections.

		Normal Wald		Expanded Welch	
n	$\|\Delta\|$	Rejections	Opposite sign	Rejections	Opposite sign
1,000	0.01	2,010	2,009	1,906	1,905
1,000	0.02	725	713	680	669
10,000	0.01	13,984	0	13,935	0
10,000	0.02	19,863	0	19,862	0

At $n = 1{,}000$ and a 0.01-nat difference, only one rejection for each method has the correct estimated direction. At 0.02 nats, 713 of 725 Wald rejections and 669 of 680 Expanded Welch rejections still have the opposite sign. The larger cutoff reduces rejection without changing any estimated difference. At $n = 10{,}000$, all rejections in these comparisons have the correct negative sign, accompanying the much smaller bias and improved power.

C.5 Four-cell construction check

Representative 4×4 comparison.

Table C.7 compares the copying and four-cell populations at matched 4×4 different-skew margins and MI targets. All outputs and diagnostics are finite and valid. The sampled populations differ between constructions, so the cross-construction comparison uses independent repetitions, whereas within each construction Wald and Expanded Welch receive the same table pairs.

Table C.7: Rejection and difference bias under two population constructions.

Construction	n	$\|\Delta\|$	Wald	Expanded	Bias
Copying	1,000	0	0.0394	0.0331	-0.00012
Copying	1,000	0.01	0.1132	0.1013	-0.00004
Copying	10,000	0	0.0481	0.0473	-0.00001
Copying	10,000	0.01	0.7962	0.7940	0.00002
Four-cell	1,000	0	0.0403	0.0345	-0.00018
Four-cell	1,000	0.01	0.1148	0.1032	-0.00018
Four-cell	10,000	0	0.0502	0.0494	-0.00001
Four-cell	10,000	0.01	0.7865	0.7844	-0.00003

The same qualitative method comparison persists: at $n = 1{,}000$, Expanded Welch lowers an already-low null rate and reduces power; at 10,000 its effect is smaller. The construction still changes the estimated power. For the 0.01-nat difference at $n = 10{,}000$, Wald's estimated power is 0.7962 under copying and 0.7865 under the four-cell transfer. The difference, 0.0098, has approximate independent-stream 95% interval [0.0018, 0.0177]. Both constructions give the same qualitative comparison between the tests at these settings, while numerical power differs.

Sparse 16×16 comparison.

Table C.8 tests the larger-table failure under the same two constructions. Both use different-skew margins $m_{16}(0.7)$ for P and $m_{16}(0.8)$ for Q, with the MI targets and sample sizes matched. Every setting has 20,000 repetitions and full validity. Rejection brackets give Wilson 95% intervals. Bias and its Monte Carlo standard error (MCSE), and observed SD, describe the corrected MI difference in nats. Est./obs. divides mean squared standard error by observed sampling variance. Opposite W/E reports the percentage of Wald and Expanded Welch rejections whose estimated difference has the opposite sign to the population difference. It applies only to the nonzero alternative.

Table C.8: Matched 16×16 comparisons under two constructions.

n	Construction	Normal Wald	Expanded Welch	Valid W/E (%)
$I(P) = I(Q) = 0.02$: false positives				
1,000	Copying	0.2522 [0.2462, 0.2583]	0.2402 [0.2343, 0.2462]	100.0/100.0
1,000	Four-cell	0.3095 [0.3031, 0.3159]	0.2977 [0.2914, 0.3041]	100.0/100.0
10,000	Copying	0.0463 [0.0434, 0.0492]	0.0453 [0.0425, 0.0482]	100.0/100.0
10,000	Four-cell	0.0499 [0.0470, 0.0530]	0.0491 [0.0462, 0.0522]	100.0/100.0
$I(P) = 0.02,\ I(Q) = 0.03$: power				
1,000	Copying	0.1005 [0.0964, 0.1047]	0.0953 [0.0913, 0.0994]	100.0/100.0
1,000	Four-cell	0.1247 [0.1202, 0.1294]	0.1176 [0.1132, 0.1221]	100.0/100.0
10,000	Copying	0.6992 [0.6928, 0.7055]	0.6968 [0.6903, 0.7031]	100.0/100.0
10,000	Four-cell	0.6323 [0.6256, 0.6390]	0.6297 [0.6229, 0.6363]	100.0/100.0

n	Construction	Bias	MCSE	Obs. SD	Est./obs.	Opposite W/E (%)
$I(P) = I(Q) = 0.02$						
1,000	Copying	+0.02960	0.00010	0.01449	1.94	—
1,000	Four-cell	+0.03145	0.00011	0.01568	1.67	—
10,000	Copying	-0.00088	0.00003	0.00387	1.07	—
10,000	Four-cell	-0.00097	0.00003	0.00426	1.05	—
$I(P) = 0.02,\ I(Q) = 0.03$						
1,000	Copying	+0.03061	0.00011	0.01564	1.81	99.95/99.95
1,000	Four-cell	+0.03136	0.00012	0.01671	1.60	99.96/99.96
10,000	Copying	-0.00094	0.00003	0.00433	1.05	0.00/0.00
10,000	Four-cell	-0.00092	0.00003	0.00468	1.04	0.00/0.00

At $n = 1{,}000$, the four-cell null has difference bias 0.03145 nats and false-positive rates 0.3095 for Wald and 0.2977 for Expanded Welch. Both rates exceed those under copying, where the corresponding bias is 0.02960 nats. For the 0.01-nat alternative, the four-cell bias of 0.03136 moves the mean estimated difference to approximately $-0.01 + 0.03136 = 0.02136$ nats. This is positive although Q has larger population MI. Opposite-sign counts are 2,493 of 2,494 Wald rejections and 2,350 of 2,351 Expanded Welch rejections; only one rejection for each method has the correct sign. The sparse-table bias and reversed estimated ordering therefore persist under both tested constructions.

At $n = 10{,}000$, the four-cell null rates are 0.0499 and 0.0491, with much smaller difference bias. Every rejection at the alternative has the correct estimated direction under either construction. Power nevertheless differs: Wald gives 0.6992 under copying and 0.6323 under the four-cell transfer. The MI difference biases under copying and the four-cell transfer are similar, but the observed SD rises from approximately 0.00433 to 0.00468 nats, making the same population difference harder to detect under the transfer. Changing association arrangement thus changes precision even when table size, margins and MI targets match.

C.6 Unequal sample sizes at a fixed total

Table C.9 compares three allocations of 2,000 observations between the same copying 4×4 populations. The row and column margins are $(0.7, 0.1, 0.1, 0.1)$ in P and $(0.8, 0.2/3, 0.2/3, 0.2/3)$ in Q. Both MI values are 0.02 nats under the null; under the alternative, $I(Q) = 0.03$ and $\Delta = -0.01$ nats. Each allocation uses 20,000 independent table pairs, and the balanced results are the corresponding main-study comparisons. Different allocations use independent repetitions, whereas the two methods receive the same samples within each allocation.

Rejection divides valid rejections by all repetitions; brackets give pointwise Wilson 95% intervals. Valid W/E gives the valid-result percentages for Wald and Expanded Welch. Difference bias and observed SD are in nats. Est./obs. compares mean squared standard error with observed sampling variance, and DF is the combined degrees of freedom. All estimates and test results are finite and valid in these comparisons.

Table C.9: Rejection and estimation diagnostics at fixed total sample size.

(n_P, n_Q)	Normal Wald	Expanded Welch	Valid W/E (%)
$I(P) = I(Q) = 0.02$: false positives			
$(1000, 1000)$	0.0394 [0.0368, 0.0422]	0.0331 [0.0307, 0.0357]	100.0/100.0
$(400, 1600)$	0.0463 [0.0435, 0.0493]	0.0306 [0.0283, 0.0331]	100.0/100.0
$(1600, 400)$	0.0454 [0.0426, 0.0484]	0.0252 [0.0231, 0.0274]	100.0/100.0
$I(P) = 0.02,\ I(Q) = 0.03$: power			
$(1000, 1000)$	0.1132 [0.1089, 0.1177]	0.1013 [0.0972, 0.1056]	100.0/100.0
$(400, 1600)$	0.1295 [0.1249, 0.1342]	0.0997 [0.0956, 0.1039]	100.0/100.0
$(1600, 400)$	0.0540 [0.0509, 0.0572]	0.0340 [0.0316, 0.0366]	100.0/100.0

(n_P, n_Q)	Difference bias	Observed SD	Est./obs.	Median DF
$I(P) = I(Q) = 0.02$				
$(1000, 1000)$	-0.00012	0.01053	1.06	35.70
$(400, 1600)$	+0.00068	0.01374	1.10	14.42
$(1600, 400)$	-0.00144	0.01376	1.11	10.97
$I(P) = 0.02,\ I(Q) = 0.03$				
$(1000, 1000)$	-0.00004	0.01167	1.07	42.10
$(400, 1600)$	+0.00073	0.01431	1.10	16.95
$(1600, 400)$	-0.00143	0.01610	1.08	13.48

With equal MI, Wald rejects at 0.0394 for balanced samples, 0.0463 at $(n_P, n_Q) = (400, 1600)$, and 0.0454 at $(1600, 400)$. Expanded Welch lowers these rates to 0.0331, 0.0306 and 0.0252, respectively. The imbalance therefore does not create excessive Wald rejection in these settings, and the larger Expanded Welch cutoff makes already-low rates lower. The null median degrees of freedom fall from 35.70 under balance to 14.42 and 10.97, increasing the cutoff from the Student t distribution.

For the 0.01-nat difference, giving Q only 400 observations lowers power for both methods. The observed SD rises from 0.01167 nats under balance to 0.01610; with P limited to 400, it is 0.01431. Reversing the allocation also changes the remaining difference bias: $+0.00073$ nats when P has fewer observations and -0.00143 when Q has fewer. The null biases have the same signs, with Monte Carlo standard error about 0.00010 in either case. Mean squared standard error exceeds observed variance in each comparison. Allocation therefore changes the estimated MI difference and standard error as well as Expanded Welch's rejection cutoff.

C.7 Holm adjustment for the application

Table C.10 gives the raw and adjusted p-values for all six comparisons used to select the online-shopping application. Each comparison tests whether predictor–outcome MI is equal between two groups. The Adult comparisons use female and male records, with income above US$50,000 as the binary outcome and the original education (16 categories) or occupation (14 categories) as the predictor (Becker and Kohavi, 1996). The original training and test partitions are combined, and records are counted without survey weights. Only the occupation comparison excludes records with missing occupation.

The Student Performance comparisons use Portuguese-language course records from schools GP and MS, with a final grade of at least 10 as the binary outcome. The predictors are the original study-time categories (four levels) and maternal-education categories (five levels) (Cortez, 2008). The shopping comparisons use new and returning visitors, with purchase as the binary outcome and month or weekend as the predictor.

Table C.10: Raw and Holm-adjusted p-values for the six exploratory comparisons.

	Normal Wald		Expanded Welch	
Comparison	Raw	Adjusted	Raw	Adjusted
Adult: occupation	1.18×10^{-44}	7.10×10^{-44}	1.64×10^{-42}	9.81×10^{-42}
Adult: education	1.89×10^{-34}	9.45×10^{-34}	4.93×10^{-33}	2.46×10^{-32}
Online Shoppers: month	4.71×10^{-5}	1.89×10^{-4}	3.73×10^{-4}	1.49×10^{-3}
Student Performance: study time	0.1982	0.5946	0.2339	0.7016
Student Performance: maternal education	0.8361	1	0.8468	1
Online Shoppers: weekend	0.9554	1	0.9601	1

Holm's multiple-testing adjustment is applied separately to each method's six p-values (Holm, 1979). Order the raw p-values from smallest to largest and multiply them by 6, 5, 4, 3, 2 and 1, respectively. Moving through that order, replace any value smaller than the preceding adjusted value by that preceding value, and cap the results at one. The month–purchase comparison has the third-smallest raw p-value for both methods, and the earlier adjusted values are smaller, so its adjusted p-value is four times its raw value. Both adjusted month p-values remain below 0.05.

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